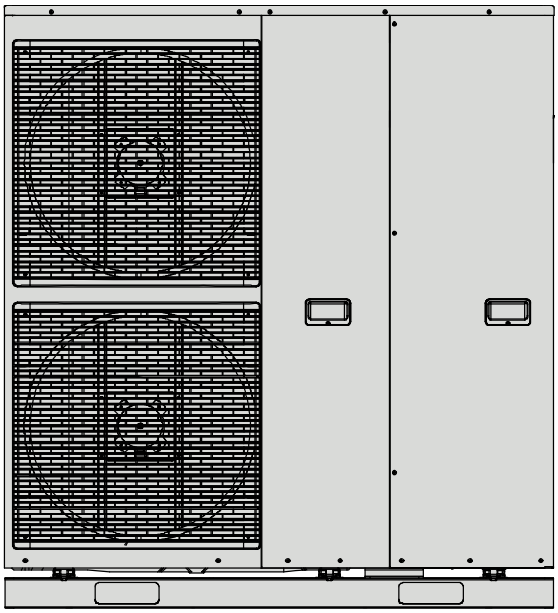
Order No: PAPAMY2101001CE

Service Manual

Air-to-Water Heatpump



Mono Bloc Unit
WH-MXC09J3E5
WH-MXC12J6E5

Destination
Europe
Turkey

⚠ WARNING

This service information is designed for experienced repair technicians only and is not designed for use by the general public.
It does not contain warnings or cautions to advise non-technical individuals of potential dangers in attempting to service a product.
Products powered by electricity should be serviced or repaired only by experienced professional technicians. Any attempt to service or repair the products dealt with in this service information by anyone else could result in serious injury or death.

⚠ CAUTION

R32 REFRIGERANT – This Air Conditioner contains and operates with refrigerant R32.
THIS PRODUCT MUST ONLY BE INSTALLED OR SERVICED BY QUALIFIED PERSONNEL.
Refer to National, State, Territory and local legislation, regulations, codes, installation & operation manuals, before the installation, maintenance and/or service of this product.

⚠ PRECAUTION OF LOW TEMPERATURE

In order to avoid frostbite, be assured of no refrigerant leakage during the installation or repairing of refrigerant circuit.

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• Specifications, designs and contents in this Service Manual are subject to change without notice.

1. Safety Precautions

- Read the following “SAFETY PRECAUTIONS” carefully before installation of (Mono bloc) Air-to-Water Heatpump system (hereafter referred to as “Mono bloc unit”).
- Electrical works and water installation works must be done by licensed electrician and licensed water system installer respectively. Be sure to use the correct rating and main circuit for the model to be installed.
- The caution items stated here must be followed because these important contents are related to safety. The meaning of each indication used is as below.
  Incorrect installation due to ignorance or negligence of the instructions will cause harm or damage, and the seriousness is classified by the following indications.

⚠ **WARNING**	This indication shows the possibility of causing death or serious injury.
⚠ **CAUTION**	This indication shows the possibility of causing injury or damage to properties only.

The items to be followed are classified by the symbols:

⃠	Symbol with white background denotes item that is PROHIBITED.
❗ ⏚	Symbol with dark background denotes item that must be carried out.

- Carry out test run to confirm that no abnormality occurs after the installation. Then, explain to user the operation, care and maintenance as stated in instructions. Please remind the customer to keep the operating instructions for future reference.
- If there is any doubt about the installation procedure or operation, always contact the authorized dealer for advice and information.
- This appliance is not intended for accessibility by the general public.
- Please leave this installation manual to the user after installation.

	⚠ **WARNING**	
1.	Do not use means to accelerate the defrosting process or to clean, other than those recommended by the manufacturer. Any unfit method or using incompatible material may cause product damage, burst and serious injury.	⃠
2.	Do not install Mono bloc unit near handrail of veranda. When installing Mono bloc unit at veranda of high rise building, child may climb up to Mono bloc unit and cross over the handrail and causing accident.	⃠
3.	Do not use unspecified cord, modified cord, joint cord or extension cord for power supply cord. Do not share the single outlet with other electrical appliances. Poor contact, poor insulation or over current will cause electrical shock or fire.	⃠
4.	Do not tie up the power supply cord into a bundle by band. Abnormal temperature rise on power supply cord may happen.	
5.	Do not insert your fingers or other objects into the unit, high speed rotating fan may cause injury.	⃠
6.	Do not sit or step on the unit, you may fall down accidentally.	⃠
7.	Keep plastic bag (packaging material) away from small children, it may cling to nose and mouth and prevent breathing.	⃠
8.	Do not purchase unauthorized electrical parts for installation, service, maintenance and etc.. They might cause electrical shock or fire.	⃠
9.	Do not pierce or burn as the appliance is pressurized. Do not expose the appliance to heat, flame, sparks, or other sources of ignition. Else, it may explode and cause injury or death.	⃠
10.	This Mono bloc unit is a multi supply appliance. All circuits must be disconnected before accessing the unit terminals.	⃠
11.	Do not modify the wiring of Mono bloc unit for installation of other components (i.e. heater, etc). Overloaded wiring or wire connection points may cause electrical shock or fire.	⃠
12.	Do not add or replace refrigerant other than specified type. It may cause product damage, burst and injury etc.	⃠
13.	For electrical work, follow the national regulation, legislation and this installation instructions. An independent circuit and single outlet must be used. If electrical circuit capacity is not enough or defect found in the electrical work, it will cause electrical shock or fire.	❗
14.	For water circuit installation work, follow to relevant European and national regulations (including EN61770) and local plumbing and building regulation codes.	❗
15.	Engage authorized dealer or specialist for installation. If installation done by the user is incorrect, it will cause water leakage, electrical shock or fire.	❗
16.	Install according to this installation instructions strictly. If installation is defective, it will cause water leakage, electrical shock or fire.	❗
17.	Only use the supplied or specified installation parts. Else, it may causes Mono bloc unit vibrate, fall, water leakage, electrical shock or fire.	❗
18.	Install at a flat, strong and firm location which is able to withstand the Mono bloc unit’s weight. If the location is slanting, or strength is not enough the set will fall and cause injury.	❗
19.	Wire routing must be properly arranged so that control board cover is fixed properly. If control board cover is not fixed perfectly, it will cause fire or electrical shock.	❗

	⚠ WARNING	
20.	This equipment is strongly recommended to be installed with Residual Current Device (RCD) on-site according to the respective national wiring rules or country-specific safety measures in terms of residual current.	
21.	The unit is only for use in a closed water system. Utilization in an open water system may lead to excessive corrosion of the water piping and risk of incubating bacteria colonies, particularly Legionella, in water.	
22.	If there is any doubt about the installation procedure or operation, always contact the authorized dealer for advice and information.	
23.	Select a location where in case of water leakage, the leakage will not cause damage to other properties.	
24.	When installing electrical equipment at wooden building of metal lath or wire lath, in accordance with electrical facility standard, no electrical contact between equipment and building is allowed. Insulator must be installed in between.	
25.	This installation may be subjected to building regulation approval applicable to respective country that may require to notify the local authority before installation.	
26.	Any work carried out on the Mono bloc unit after removing any panel which is secured by screws, must be carried out under the supervision of authorized dealer and licensed installation contractor.	
27.	Be aware that refrigerants may not contain an odour.	
28.	This equipment must be properly earthed. Earth line must not be connected to gas pipe, water pipe, earth of lightning rod and telephone. Otherwise, it may cause electrical shock in case of equipment breakdown or insulation breakdown.	

	⚠ CAUTION	
1.	Do not install the Mono bloc unit at place where leakage of flammable gas may occur. In case gas leaks and accumulates at surrounding of the Mono bloc unit, it may cause fire.	
2.	Do not release refrigerant during piping work for installation, re-installation and during repairing a refrigeration parts. Take care of the liquid refrigerant, it may cause frostbite.	
3.	Make sure the insulation of power supply cord does not contact hot part (i.e. refrigerant piping, water piping) to prevent from insulation failure (melt).	
4.	Do not touch the sharp aluminium fin, sharp parts may cause injury.	
5.	Do not apply excessive force to water pipes that may damage the pipes. If water leakage occurs, it will cause flooding and damage to other properties.	
6.	Carry out drainage piping as mentioned in installation instructions. If drainage is not perfect, water leakage may happen and may cause damage to properties of the user.	
7.	The piping installation work must be flushed before the Mono bloc unit is connected to remove contaminants. Contaminants may damage the Mono bloc unit components.	
8.	Select an installation location which is easy for maintenance. Incorrect installation, service or repair of this Mono bloc unit may increase the risk of rupture and this may result in loss damage or injury and/or property.	
9.	Power supply connection to Mono bloc unit. • Power supply point should be in easily accessible place for power disconnection in case of emergency. • Must follow local national wiring standard, regulation and this installation instruction. • Strongly recommended to make permanent connection to a circuit breaker. It must be a double pole switch with a minimum 3.0 mm gap. - Power supply 1: Use approved 30A circuit breaker - Power supply 2: Use approved 30A circuit breaker	
10.	Ensure the correct polarity is maintained throughout all wiring. Otherwise, it will cause electrical shock or fire.	
11.	After installation, the installer is obliged to verify correct operation of the Mono bloc unit. Check the connection point for water leakage during test run. If leakage occurs, it will cause damage to other properties.	
12.	Keep any required ventilation openings clear of obstruction.	
13.	Installation work. Four or more people are required to carry out the installation work. The weight of Mono bloc unit might cause injury if carried by less than four people.	

2. Precaution for Using R32 Refrigerant

- The basic installation work procedures are the same as conventional refrigerant (R410A, R22) models. However, pay careful attention to the following points:

	⚠ WARNING	
1.	Since the working pressure is higher than that of refrigerant R22 models, some of the piping and installation and service tools are special. Especially, when replacing a refrigerant R22 model with a new refrigerant R32 model, always replace the conventional piping with the R32 and R410A piping on the outdoor unit side. For R32 and R410A, the same pipe can be used.	❗
2.	The mixing of different refrigerants within a system is prohibited. Models that use refrigerant R32 and R410A have a different charging port thread diameter to prevent erroneous charging with refrigerant R22 and for safety. Therefore, check beforehand. [The charging port thread diameter for R32 and R410A is 12.7 mm (1/2 inch).]	❗
3.	Ensure that foreign matter (oil, water, etc.) does not enter the piping.	❗
4.	Operation, maintenance, repairing and refrigerant recovery should be carried out by trained and certified personnel in the use of flammable refrigerants and as recommended by the manufacturer. Any personnel conducting an operation, servicing or maintenance on a system or associated parts of the equipment should be trained and certified.	❗
5.	Any part of refrigerating circuit (evaporators, air coolers, AHU, condensers or liquid receivers) or piping should not be located in the proximity of heat sources, open flames, operating gas appliance or an operating electric heater.	❗
6.	The user/owner or their authorized representative shall regularly check the alarms, mechanical ventilation and detectors, at least once a year, where as required by national regulations, to ensure their correct functioning.	❗
7.	A logbook shall be maintained. The results of these checks shall be recorded in the logbook.	❗
8.	In case of ventilations in occupied spaces shall be checked to confirm no obstruction.	❗
9.	Before a new refrigerating system is put into service, the person responsible for placing the system in operation should ensure that trained and certified operating personnel are instructed on the basis of the instruction manual about the construction, supervision, operation and maintenance of the refrigerating system, as well as the safety measures to be observed, and the properties and handling of the refrigerant used.	❗
10.	The general requirement of trained and certified personnel are indicated as below: a) Knowledge of legislation, regulations and standards relating to flammable refrigerants; and, b) Detailed knowledge of and skills in handling flammable refrigerants, personal protective equipment, refrigerant leakage prevention, handling of cylinders, charging, leak detection, recovery and disposal; and, c) Able to understand and to apply in practice the requirements in the national legislation, regulations and Standards; and, d) Continuously undergo regular and further training to maintain this expertise.	❗
11.	Protect the refrigerating system from accidental rupture due to moving furniture or reconstruction activities.	❗
12.	To ensure no leaking, refrigerant joints shall be tightness tested. The test method shall have a sensitivity of 5 grams per year of refrigerant or better under a pressure of at least 0,25 times the maximum allowable pressure (>1.04MPa, max 4.3MPa). No leak shall be detected.	❗

⚠ CAUTION

1. Installation (Space)
   - Must comply with national gas regulations, state municipal rules and legislation. Notify relevant authorities in accordance with all applicable regulations.
   - Must ensure mechanical connections be accessible for maintenance purposes.
   - In cases that require mechanical ventilation, ventilation openings shall be kept clear of obstruction.
   - When disposal of the product, do follow to the precautions in #12 and comply with national regulations. Always contact to local municipal offices for proper handling.

2. Servicing

2-1. Service personnel
- Any qualified person who is involved with working on or breaking into a refrigerant circuit should hold a current valid certificate from an industry-accredited assessment authority, which authorizes their competence to handle refrigerants safely in accordance with an industry recognized assessment specification.
- Servicing shall only be performed as recommended by the equipment manufacturer. Maintenance and repair requiring the assistance of other skilled personnel shall be carried out under the supervision of the person competent in the use of flammable refrigerants.
- Servicing shall be performed only as recommended by the manufacturer.
- The system is inspected, regularly supervised and maintained by a trained and certified service personnel who is employed by the person user or party responsible.
- Ensure refrigerant charge not to leak.

2-2. Work
- Prior to beginning work on systems containing flammable refrigerants, safety checks are necessary to ensure that the risk of ignition is minimised.
  For repair to the refrigerating system, the precautions in #2-2 to #2-8 must be followed before conducting work on the system.
- Work shall be undertaken under a controlled procedure so as to minimize the risk of a flammable gas or vapour being present while the work is being performed.
- All maintenance staff and others working in the local area shall be instructed and supervised on the nature of work being carried out.
- Avoid working in confined spaces. Always ensure away from source, at least 2 meter of safety distance, or zoning of free space area of at least 2 meter in radius.
- Wear appropriate protective equipment, including respiratory protection, as conditions warrant.
- Keep all sources of ignition and hot metal surfaces away.

2-3. Checking for presence of refrigerant
- The area shall be checked with an appropriate refrigerant detector prior to and during work, to ensure the technician is aware of potentially flammable atmospheres.
- Ensure that the leak detection equipment being used is suitable for use with flammable refrigerants, i.e. non sparking, adequately sealed or intrinsically safe.
- In case of leakage/spillage happened, immediately ventilate area and stay upwind and away from spill/release.
- In case of leakage/spillage happened, do notify persons downwind of the leaking/spill, isolate immediate hazard area and keep unauthorized personnel out.

2-4. Presence of fire extinguisher
- If any hot work is to be conducted on the refrigeration equipment or any associated parts, appropriate fire extinguishing equipment shall be available at hand.
- Have a dry powder or CO_2 fire extinguisher adjacent to the charging area.

2-5. No ignition sources
- No person carrying out work in relation to a refrigerating system which involves exposing any pipe work that contains or has contained flammable refrigerant shall use any sources of ignition in such a manner that it may lead to the risk of fire or explosion. He/She must not be smoking when carrying out such work.
- All possible ignition sources, including cigarette smoking, should be kept sufficiently far away from the site of installation, repairing, removing and disposal, during which flammable refrigerant can possibly be released to the surrounding space.
- Prior to work taking place, the area around the equipment is to be surveyed to make sure that there are no flammable hazards or ignition risks.
- “No Smoking” signs shall be displayed.

2-6. Ventilated area
- Ensure that the area is in the open or that it is adequately ventilated before breaking into the system or conducting any hot work.
- A degree of ventilation shall continue during the period that the work is carried out.
- The ventilation should safely disperse any released refrigerant and preferably expel it externally into the atmosphere.

2-7. Checks to the refrigerating equipment
- Where electrical components are being changed, they shall be fit for the purpose and to the correct specification.
- At all times the manufacturer’s maintenance and service guidelines shall be followed.
- If in doubt consult the manufacturer’s technical department for assistance.
- The following checks shall be applied to installations using flammable refrigerants.
  - The ventilation machinery and outlets are operating adequately and are not obstructed.
  - Marking to the equipment continues to be visible and legible. Markings and signs that are illegible shall be corrected.

2-8. Checks to electrical devices
- Repair and maintenance to electrical components shall include initial safety checks and component inspection procedures.
- Initial safety checks shall include but not limit to:-
  - That capacitors are discharged: this shall be done in a safe manner to avoid possibility of sparking.
  - That there is no live electrical components and wiring are exposed while charging, recovering or purging the system.
  - That there is continuity of earth bonding.
- At all times the manufacturer’s maintenance and service guidelines shall be followed.
- If in doubt consult the manufacturer’s technical department for assistance.
- If a fault exists that could compromise safety, then no electrical supply shall be connected to the circuit until it is satisfactorily dealt with.
- If the fault cannot be corrected immediately but it is necessary to continue operation, an adequate temporary solution shall be used.
- The owner of the equipment must be informed or reported so all parties are advised thereinafter.

⚠ CAUTION

3. Repairs to sealed components

- During repairs to sealed components, all electrical supplies shall be disconnected from the equipment being worked upon prior to any removal of sealed covers, etc.
- If it is absolutely necessary to have an electrical supply to equipment during servicing, then a permanently operating form of leak detection shall be located at the most critical point to warn of a potentially hazardous situation.
- Particular attention shall be paid to the following to ensure that by working on electrical components, the casing is not altered in such a way that the level of protection is affected. This shall include damage to cables, excessive number of connections, terminals not made to original specification, damage to seals, incorrect fitting of glands, etc.
- Ensure that apparatus is mounted securely.
- Ensure that seals or sealing materials have not degraded such that they no longer serve the purpose of preventing the ingress of flammable atmospheres.
- Replacement parts shall be in accordance with the manufacturer's specifications.

NOTE:	The use of silicon sealant may inhibit the effectiveness of some types of leak detection equipment. Intrinsically safe components do not have to be isolated prior to working on them.

4. Repair to intrinsically safe components

- Do not apply any permanent inductive or capacitance loads to the circuit without ensuring that this will not exceed the permissible voltage and current permitted for the equipment in use.
- Intrinsically safe components are the only types that can be worked on while live in the presence of a flammable atmosphere.
- The test apparatus shall be at the correct rating.
- Replace components only with parts specified by the manufacturer. Unspecified parts by manufacturer may result ignition of refrigerant in the atmosphere from a leak.

5. Cabling

- Check that cabling will not be subject to wear, corrosion, excessive pressure, vibration, sharp edges or any other adverse environmental effects.
- The check shall also take into account the effects of aging or continual vibration from sources such as compressors or fans.

6. Detection of flammable refrigerants

- Under no circumstances shall potential sources of ignition be used in the searching or detection of refrigerant leaks.
- A halide torch (or any other detector using a naked flame) shall not be used.

7. The following leak detection methods are deemed acceptable for all refrigerant systems.

- No leaks shall be detected when using detection equipment with a sensitivity of 5 grams per year of refrigerant or better under a pressure of at least 0,25 times the maximum allowable pressure (>1.04MPa, max 4.3MPa) for example, a universal sniffer.
- Electronic leak detectors may be used to detect flammable refrigerants, but the sensitivity may not be adequate, or may need re-calibration.
  (Detection equipment shall be calibrated in a refrigerant-free area.)
- Ensure that the detector is not a potential source of ignition and is suitable for the refrigerant used.
- Leak detection equipment shall be set at a percentage of the LFL of the refrigerant and shall be calibrated to the refrigerant employed and the appropriate percentage of gas (25 % maximum) is confirmed.
- Leak detection fluids are also suitable for use with most refrigerants, for example, bubble method and fluorescent method agents. The use of detergents containing chlorine shall be avoided as the chlorine may react with the refrigerant and corrode the copper pipe-work.
- If a leak is suspected, all naked flames shall be removed/extinguished.
- If a leakage of refrigerant is found which requires brazing, all of the refrigerant shall be recovered from the system, or isolated (by means of shut off valves) in a part of the system remote from the leak. The precautions in #8 must be followed to remove the refrigerant.

8. Removal and evacuation

- When breaking into the refrigerant circuit to make repairs – or for any other purpose – conventional procedures shall be used. However, it is important that best practice is followed since flammability is a consideration.
  The following procedure shall be adhered to:

> • remove refrigerant -> • purge the circuit with inert gas -> • evacuate -> • purge again with inert gas ->
> • open the circuit by cutting or brazing

- The refrigerant charge shall be recovered into the correct recovery cylinders.
- The system shall be purged with OFN to render the appliances safe. (remark: OFN = oxygen free nitrogen, type of inert gas)
- This process may need to be repeated several times.
- Compressed air or oxygen shall not be used for this task.
- Purging shall be achieved by breaking the vacuum in the system with OFN and continuing to fill until the working pressure is achieved, then venting to atmosphere, and finally pulling down to a vacuum.
- This process shall be repeated until no refrigerant is within the system.
- When the final OFN charge is used, the system shall be vented down to atmospheric pressure to enable work to take place.
- This operation is absolutely vital if brazing operations on the pipe work are to take place.
- Ensure that the outlet for the vacuum pump is not close to any ignition sources and there is ventilation available.

⚠ CAUTION

9.

Charging procedures

- In addition to conventional charging procedures, the following requirements shall be followed.
  - Ensure that contamination of different refrigerants does not occur when using charging equipment.
  - Hoses or lines shall be as short as possible to minimize the amount of refrigerant contained in them.
  - Cylinders shall be kept in an appropriate position according to the instructions.
  - Ensure that the refrigerating system is earthed prior to charging the system with refrigerant.
  - Label the system when charging is complete (if not already).
  - Extreme care shall be taken not to over fill the refrigeration system.
- Prior to recharging the system it shall be pressure tested with OFN (refer to #7).
- The system shall be leak tested on completion of charging but prior to commissioning.
- A follow up leak test shall be carried out prior to leaving the site.
- Electrostatic charge may accumulate and create a hazardous condition when charging and discharging the refrigerant. To avoid fire or explosion, dissipate static electricity during transfer by grounding and bonding containers and equipment before charging/discharging.

10.

Decommissioning

- Before carrying out this procedure, it is essential that the technician is completely familiar with the equipment and all its details.
- It is recommended good practice that all refrigerants are recovered safely.
- Prior to the task being carried out, an oil and refrigerant sample shall be taken in case analysis is required prior to re-use of reclaimed refrigerant.
- It is essential that electrical power is available before the task is commenced.
  a) Become familiar with the equipment and its operation.
  b) Isolate system electrically.
  c) Before attempting the procedure ensure that:

  • mechanical handling equipment is available, if required, for handling refrigerant cylinders;
  • all personal protective equipment is available and being used correctly;
  • the recovery process is supervised at all times by a competent person;
  • recovery equipment and cylinders conform to the appropriate standards.

  d) Pump down refrigerant system, if possible.
  e) If a vacuum is not possible, make a manifold so that refrigerant can be removed from various parts of the system.
  f) Make sure that cylinder is situated on the scales before recovery takes place.
  g) Start the recovery machine and operate in accordance with manufacturer's instructions.
  h) Do not over fill cylinders. (No more than 80 % volume liquid charge).
  i) Do not exceed the maximum working pressure of the cylinder, even temporarily.
  j) When the cylinders have been filled correctly and the process completed, make sure that the cylinders and the equipment are removed from site promptly and all isolation valves on the equipment are closed off.
  k) Recovered refrigerant shall not be charged into another refrigeration system unless it has been cleaned and checked.
- Electrostatic charge may accumulate and create a hazardous condition when charging or discharging the refrigerant. To avoid fire or explosion, dissipate static electricity during transfer by grounding and bonding containers and equipment before charging/discharging.

11.

Labelling

- Equipment shall be labelled stating that it has been de-commissioned and emptied of refrigerant.
- The label shall be dated and signed.
- Ensure that there are labels on the equipment stating the equipment contains flammable refrigerant.

12.

Recovery

- When removing refrigerant from a system, either for servicing or decommissioning, it is recommended good practice that all refrigerants are removed safely.
- When transferring refrigerant into cylinders, ensure that only appropriate refrigerant recovery cylinders are employed.
- Ensure that the correct number of cylinders for holding the total system charge are available.
- All cylinders to be used are designated for the recovered refrigerant and labelled for that refrigerant (i.e. special cylinders for the recovery of refrigerant).
- Cylinders shall be complete with pressure relief valve and associated shut-off valves in good working order.
- Recovery cylinders are evacuated and, if possible, cooled before recovery occurs.
- The recovery equipment shall be in good working order with a set of instructions concerning the equipment that is at hand and shall be suitable for the recovery of concerning the equipment that is at hand and shall be suitable for the recovery of flammable refrigerants.
- In addition, a set of calibrated weighing scales shall be available and in good working order.
- Hoses shall be complete with leak-free disconnect couplings and in good condition.
- Before using the recovery machine, check that it is in satisfactory working order, has been properly maintained and that any associated electrical components are sealed to prevent ignition in the event of a refrigerant release. Consult manufacturer if in doubt.
- The recovered refrigerant shall be returned to the refrigerant supplier in the correct recovery cylinder, and the relevant Waste Transfer Note arranged.
- Do not mix refrigerants in recovery units and especially not in cylinders.
- If compressors or compressor oils are to be removed, ensure that they have been evacuated to an acceptable level to make certain that flammable refrigerant does not remain within the lubricant.
- The evacuation process shall be carried out prior to returning the compressor to the suppliers.
- Only electric heating to the compressor body shall be employed to accelerate this process.
- When oil is drained from a system, it shall be carried out safely.

3. Specifications

3.1 WH-MXC09J3E5

Item			Unit	Refrigerant System		
Performance Test Condition				EN 14511 / EN14825		
POS (EAN)				5025232910687		
Cooling Capacity			Condition (Ambient/Water)	A35W7		
			kW	9.00		
			BTU/h	30700		
			kcal/h	7740		
Cooling EER			W/W	3.18		
			kcal/h	2.73		
Heating Capacity			Condition (Ambient/Water)	A7W35	A2W35	
			kW	9.00	9.00	
			BTU/h	30700	30700	
			kcal/h	7740	7740	
Heating COP			W/W	5.08	3.81	
			kcal/h	4.37	3.28	
Heating ErP	Low temperature Application (W35)					
	Application		Climate	Warmer	Average	Colder
	Pdesign		kW	9.0	9.0	11.0
	Tbivalent / TOL		°C	2 / 2	-10 / -10	-15 / -22
	SCOP / ns		(W/W) / %	6.47 / 256	4.96 / 195	4.31 / 169
	Annual Consumption		kWh	1859	3747	6289
	Class			A+++	A+++	A++
	Low temperature Application (W55)					
	Application		Climate	Warmer	Average	Colder
	Pdesign		kW	9.0	9.0	11.0
	Tbivalent / TOL		°C	2 / 2	-10 / -10	-15 / -22
	SCOP / ns		(W/W) / %	4.34 / 171	3.57 / 140	3.26 / 127
	Annual Consumption		kWh	2772	5208	8327
	Class			A+++	A++	A++
Air Flow			m^3/min (ft^3/min)	Cooling: 96.4 (3400) Heating: 76.8 (2710)		
Refrigerant Control Device				Expansion Valve		
Refrigerant Oil			cm^3	FW50S (1200)		
Refrigerant (R32)			kg (oz)	1.6 (56.4)		
F-GAS	GWP			675		
	CO2eq (ton) (Precharged / Maximum)			1.080 / -		
Compressor	Type			Hermetic Motor		
	Motor Type			Brushless (4-poles)		
	Rated Output		kW	3.00		

Item		Unit	Refrigerant System
Fan	Type		Propeller Fan
	Material		PP
	Motor Type		DC (8-poles)
	Output Power	W	60
	Fan Speed	rpm	Cooling: Top: 640, Bottom: 680 Heating: Top: 490, Bottom: 530
Heat Exchanger	Fin Material		Aluminium (Pre Coat)
	Fin Type		Corrugated Fin
	Row x Stage x FPI		2 x 51 x 19
	Size (W x H X L)	mm	38.1 x 1259.4 x 903.7:873.8

Item		Unit	Mono bloc Unit		
Dimension	Height	mm (inch)	1410 (55-17/32)		
	Width	mm (inch)	1283 (50-17/32)		
	Depth	mm (inch)	320 (12-5/8)		
Net Weight		kg (lbs)	140 (309)		
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB(A)	Cooling: 50	Heating: 51	Heating: 51
		Power level dB	Cooling: 68	Heating: 68 Heating: 65	Heating: 68 Heating: 65
Power Source (Phase, Voltage, Cycle)		Ø	Single		
		V	230		
		Hz	50		
Input Power		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		kW	Cooling: 2.83	Heating: 1.77	Heating: 2.36
Maximum Input Power for Heatpump System		kW	6.27		
Power Supply 1 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 29.0 / 6.27k		
Power Supply 2 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 3.00k		
Power Supply 3 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			- / - / -		
Starting current		A	13.1		
Running Current		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		A	Cooling: 13.1	Heating: 8.3	Heating: 10.9
Maximum Current for Heatpump System		A	29.0		
Power Factor Power factor means total figure of compressor and outdoor fan motor.		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		%	Cooling: 94	Heating: 93	Heating: 94
Power Cord	Number of core		-		
	Length	m (ft)	-		
Thermostat			Electronic Control		
Protection Device			Electronic Control		

Item		Unit	Water System
Performance Test Condition			EN14511 / EN14825
POS (EAN)			5025232910687
Operation Range	Outdoor Ambient	°C (min. / max.)	Cooling: 10 / 43 Heating: -20 / 35
	Water Outlet	°C (min. / max.)	Cooling: 5 / 20 Heating: 20/55 (Below Ambient -20°C) *3 20/60 (Ambient -15~0°C) or Above Ambient 25°C) *3 20/65 (Ambient 5~20°C) *3*4
Internal Pressure Differential		kPa	Cooling: 35 Heating: 35
Water Pipe Diameter	Inlet	inch	(1-1/4)
	Outlet	inch	(1-1/4)
Water Drain Hose Inner Diameter		mm (inch)	15.00 (19/32)
Pump	Motor Type		DC Motor
	Input Power	W	173
Hot Water Coil	Type		Brazed Plate
	No. of Plates		38
	Size (H x W x L)	mm	71.4 x 121 x 333
	Water Flow Rate	l/min (m³/h)	Cooling: 25.8 (1.5) Heating: 25.8 (1.5)
Pressure Relief Valve Water Circuit		kPa	Open: 800, Close: 640 and below
Flow Switch	Type		Electronic Sensor
Protection Device		A	Residual Current Circuit Breaker ()
Expansion Vessel	Volume	l	10
	MWP	bar	3
Capacity of Integrated Electric Heater / OLP TEMP		kW / °C	3.00

Note:

- Cooling capacities are based on outdoor air temperature of 35°C Dry Bulb with controlled indoor water inlet temperature of 12°C and water outlet temperature of 7°C.
- Heating capacities are based on outdoor air temperature of 7°C Dry Bulb (44.6°F Dry Bulb), 6°C Wet Bulb (42.8°F Wet Bulb) with controlled indoor water inlet temperature of 30°C and water outlet temperature of 35°C.
- Specification are subjected to change without prior notice for further improvement.
- Flow rate indicated are based on nominal capacity adjustment of leaving water temperature (LWT) 35°C and =5°C

*3 Between outdoor ambient -15°C and -20°C, the water outlet temperature gradually decreases from 60°C to 55°C.
*3 Between outdoor ambient 5°C and 0°C, the water outlet temperature gradually decreases from 65°C to 60°C.
*3 Between outdoor ambient 20°C and 25°C, the water outlet temperature gradually decreases from 65°C to 60°C.
*4 Only when ΔT is set to 15°C, the set temperature above 60°C will take effect.

3.2 WH-MXC12J6E5

Item		Unit	Refrigerant System		
Performance Test Condition			EN 14511 / EN14825		
POS (EAN)			5025232910694		
Cooling Capacity		Condition (Ambient/Water)	A35W7		
		kW	12.00		
		BTU/h	41000		
		kcal/h	10320		
Cooling EER		W/W	2.90		
		kcal/h	2.49		
Heating Capacity		Condition (Ambient/Water)	A7W35	A2W35	
		kW	12.00	12.00	
		BTU/h	41000	41000	
		kcal/h	10320	10320	
Heating COP		W/W	4.80	3.53	
		kcal/h	4.13	3.04	
Heating ErP	Low temperature Application (W35)				
	Application	Climate	Warmer	Average	Colder
	Pdesign	kW	9.0	9.0	11.0
	Tbivalent / TOL	°C	2 / 2	-10 / -10	-15 / -22
	SCOP / ns	(W/W) / %	6.47 / 256	4.96 / 195	4.31 / 169
	Annual Consumption	kWh	1859	3747	6289
	Class		A+++	A+++	A++
	Low temperature Application (W55)				
	Application	Climate	Warmer	Average	Colder
	Pdesign	kW	9.0	9.0	11.0
	Tbivalent / TOL	°C	2 / 2	-10 / -10	-15 / -22
	SCOP / ns	(W/W) / %	4.34 / 171	3.57 / 140	3.26 / 127
	Annual Consumption	kWh	2772	5208	8327
	Class		A+++	A++	A++
Air Flow		m^3/min (ft^3/min)	Cooling: 102.4 (3620) Heating: 81.3 (2870)		
Refrigerant Control Device			Expansion Valve		
Refrigerant Oil		cm^3	FW50S (1200)		
Refrigerant (R32)		kg (oz)	1.60 (56.4)		
F-GAS	GWP		675		
	CO2eq (ton) (Precharged / Maximum)		1.080 / -		
Compressor	Type		Hermetic Motor		
	Motor Type		Brushless (4-poles)		
	Rated Output	kW	3.00		

Item		Unit	Refrigerant System
Fan	Type		Propeller Fan
	Material		PP
	Motor Type		DC (8-poles)
	Output Power	W	60
	Fan Speed	rpm	Cooling: Top: 680, Bottom: 720 Heating: Top: 520, Bottom: 560
Heat Exchanger	Fin Material		Aluminium (Pre Coat)
	Fin Type		Corrugated Fin
	Row x Stage x FPI		2 x 51 x 19
	Size (W x H X L)	mm	38.1 x 1259.4 x 903.7 : 873.8

Item		Unit	Mono bloc Unit		
Dimension	Height	mm (inch)	1410 (55-17/32)		
	Width	mm (inch)	1283 (50-17/32)		
	Depth	mm (inch)	320 (12-5/8)		
Net Weight		kg (lbs)	140 (309)		
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB(A)	Cooling: 52	Heating: 52	Heating: 52
		Power level dB	Cooling: 69	Heating: 69 Heating: 65	Heating: 69 Heating: 65
Power Source (Phase, Voltage, Cycle)		Ø	Single		
		V	230		
		Hz	50		
Input Power		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		kW	Cooling: 4.14	Heating: 2.50	Heating: 3.40
Maximum Input Power for Heatpump System		kW	6.27		
Power Supply 1 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 29.0 / 6.27k		
Power Supply 2 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 26.0 / 6.00k		
Power Supply 3 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			- / - / -		
Starting current		A	19.1		
Running Current		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		A	Cooling: 19.1	Heating: 11.6	Heating: 15.7
Maximum Current for Heatpump System		A	29.0		
Power Factor Power factor means total figure of compressor and outdoor fan motor.		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		%	Cooling: 94	Heating: 94	Heating: 94
Power Cord	Number of core		-		
	Length	m (ft)	-		
Thermostat			Electronic Control		
Protection Device			Electronic Control		

Item		Unit	Water System
Performance Test Condition			EN14511 / EN14825
POS (EAN)			5025232910694
Operation Range	Outdoor Ambient	°C (min. / max.)	Cooling: 10 / 43 Heating: -20 / 35
	Water Outlet	°C (min. / max.)	Cooling: 5 / 20 Heating: 20/55 (Below Ambient -20°C) *3 20/60 (Ambient -15~0°C) or Above Ambient 25°C) *3 20/65 (Ambient 5~20°C) *3*4
Internal Pressure Differential		kPa	Cooling: 55 Heating: 55
Water Pipe Diameter	Inlet	inch	(1-1/4)
	Outlet	inch	(1-1/4)
Water Drain Hose Inner Diameter		mm (inch)	15.00 (19/32)
Pump	Motor Type		DC Motor
	Input Power	W	173
Hot Water Coil	Type		Brazed Plate
	No. of Plates		38
	Size (H x W x L)	mm	71.4 x 121 x 333
	Water Flow Rate	l/min (m³/h)	Cooling: 34.4 (2.1) Heating: 34.4 (2.1)
Pressure Relief Valve Water Circuit		kPa	Open: 800, Close: 640 and below
Flow Switch	Type		Electronic Sensor
Protection Device		A	Residual Current Circuit Breaker ()
Expansion Vessel	Volume	l	10
	MWP	bar	3
Capacity of Integrated Electric Heater / OLP TEMP		kW / °C	6.00

Note:

- Cooling capacities are based on outdoor air temperature of 35°C Dry Bulb with controlled indoor water inlet temperature of 12°C and water outlet temperature of 7°C.
- Heating capacities are based on outdoor air temperature of 7°C Dry Bulb (44.6°F Dry Bulb), 6°C Wet Bulb (42.8°F Wet Bulb) with controlled indoor water inlet temperature of 30°C and water outlet temperature of 35°C.
- Specification are subjected to change without prior notice for further improvement.
- Flow rate indicated are based on nominal capacity adjustment of leaving water temperature (LWT) 35°C and =5°C

*3 Between outdoor ambient -15°C and -20°C, the water outlet temperature gradually decreases from 60°C to 55°C.
*3 Between outdoor ambient 5°C and 0°C, the water outlet temperature gradually decreases from 65°C to 60°C.
*3 Between outdoor ambient 20°C and 25°C, the water outlet temperature gradually decreases from 65°C to 60°C.
*4 Only when ΔT is set to 15°C, the set temperature above 60°C will take effect.

4. Features

- **Inverter Technology**
  - Energy saving
- **High Efficiency**
- **Compact Design**
- **Environment Protection**
  - Non-ozone depletion substances refrigerant (R32)
- **Easy to use control panel**
- **Weekly Timer setting**
- **Quality Improvement**
  - Random auto restart after power failure for safety restart operation
  - Gas leakage protection
  - Prevent compressor reverse cycle
  - Inner protector to protect compressor
- **Serviceability Improvement**
  - Breakdown Self Diagnosis function
  - System Status Check Buttons for servicing purpose
  - System Pumpdown Button for servicing purpose
  - Front maintenance design for outdoor unit
- **Operation Condition**

	HEATING (CIRCUIT)	[*1][*2]COOLING (CIRCUIT)
Water outlet temperature (°C) (Min. / Max)	20 / 55 (Below Ambient -20°C) [*3] 20 / 60 (Ambient -15 ~ 0°C or Above Ambient 25°C) *3 20 / 65 (Ambient 5 ~ 20°C) [*3][*4]	5 / 20
Outdoor ambient temperature (°C) (Min. / Max)	-20 / 35	10 / 43

NOTICE : When the outdoor temperature is out of the above temperature range, the heating capacity will drop significantly and outdoor unit might stop for protection control.

[*3] Between outdoor ambient -15°C and -20°C, the water outlet temperature gradually decreases from 60°C to 55°C.

[*3] Between outdoor ambient 5°C and 0°C, the water outlet temperature gradually decreases from 65°C to 60°C.

[*3] Between outdoor ambient 20°C and 25°C, the water outlet temperature gradually decreases from 65°C to 60°C.

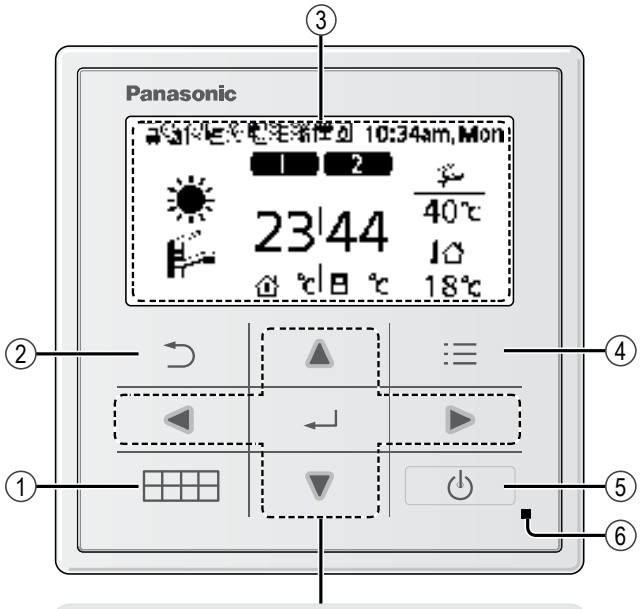
[*4] Only when ΔT is set to 15°C, the set temperature above 60°C will take effect.

5. Location of Controls and Components

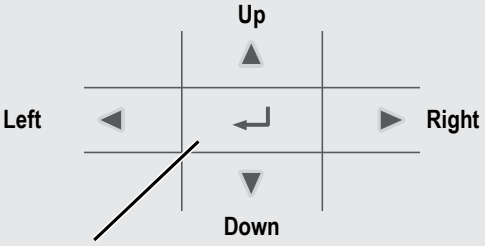
5.1 Indoor Unit

5.1.1 Remote Controller buttons and display

Buttons / Indicator	
①	**Quick Menu button** (For more details, refer to the separate Quick Menu Guide.)
②	**Back button** Returns to the previous screen
③	**LCD Display**
④	**Main Menu button** For function setup
⑤	**ON/OFF button** Starts/Stops operation
⑥	**Operation indicator** Illuminates during operation, blinks during alarm.

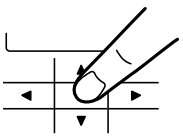


Cross key buttons
Selects an item.

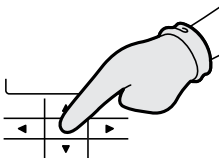


Enter button
Fixes the selected content.

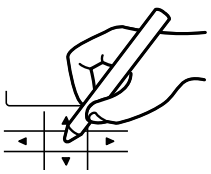
Press centre

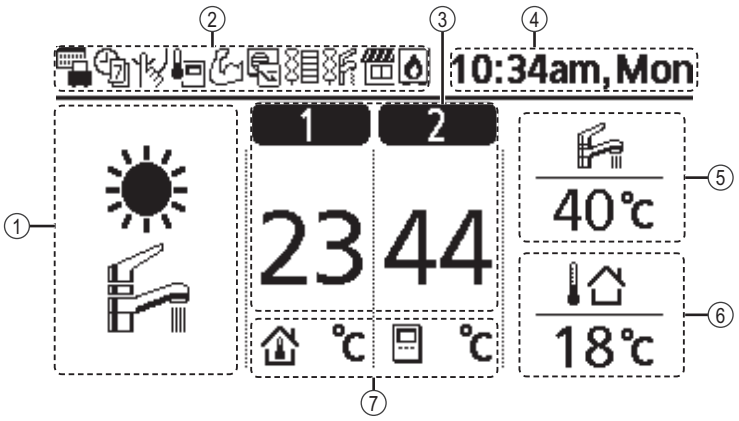


No glove



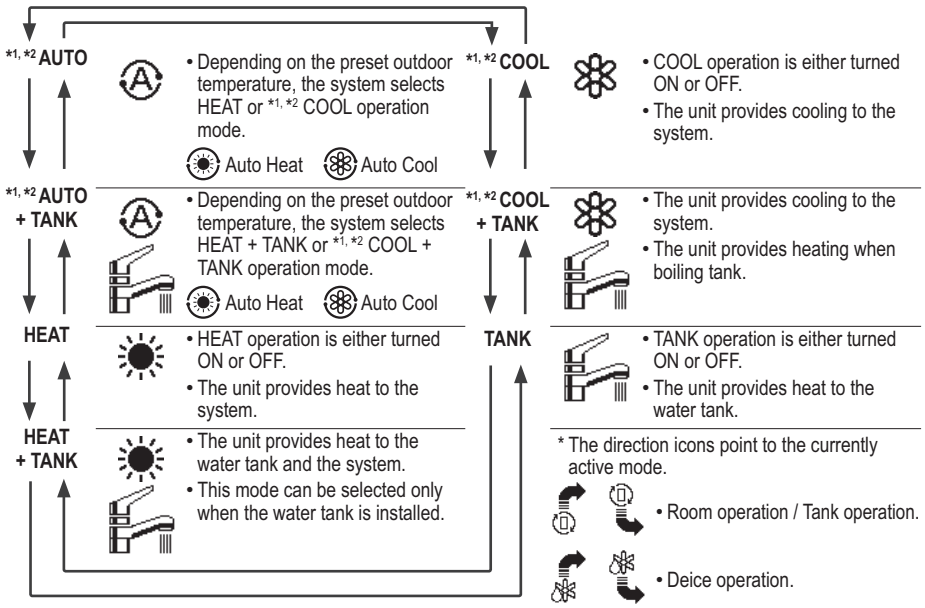
No pen





Display

① **Mode selection**



② **Operation icons**

The status of operation is displayed.
Icon will not display (under operation OFF screen) whenever operation is OFF except weekly timer.

Holiday operation status	Weekly Timer operation status	Quiet operation status
Zone:Room Thermostat →Internal sensor status	Powerful operation status	Demand Control or SG ready or SHP status
Room Heater status	Tank Heater status	Solar status
Bivalent status (Boiler)		

*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

③ **Temperature of each zone**

④ **Time and day**

⑤ **Water Tank temperature**

⑥ **Outdoor temperature**

⑦ **Sensor type/Set temperature type icons**

Water Temperature →Compensation curve	Water Temperature →Direct	Pool only
Room Thermostat →External	Room Thermostat →Internal	

5.1.2 Initialization

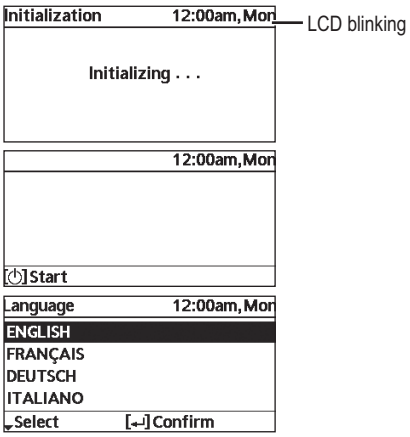
Before starting to install the various menu settings, please initiate the Remote Controller by selecting the language of operation and installing the date and time correctly.
When power is turned on for the first time, it becomes the setting screen automatically. It can also be set from personal setting of the menu.

Selecting the language

Wait while the display is initializing.
When initializing screen ends, it turns to normal screen.
When any button is pressed, language setting screen appears.

① **Scroll with ▼ and ▲ to select the language.**

② **Press ↵ to confirm the selection.**



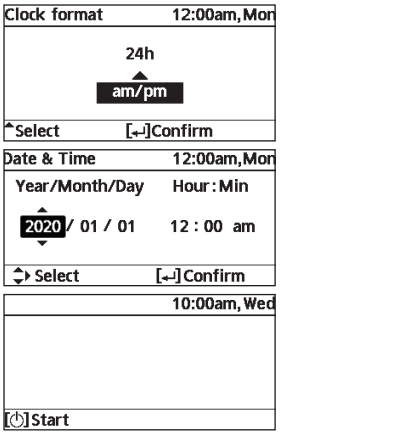
Setting the clock

① **Select with ▼ or ▲ how to display the time, either 24h or am/pm format (for example, 15:00 or 3:00 pm).**

② **Press ↵ to confirm the selection.**

③ **Use ▼ and ▲ to select year, month, day, hour and minutes. (Select and move with ▶ and press ↵ to confirm.)**

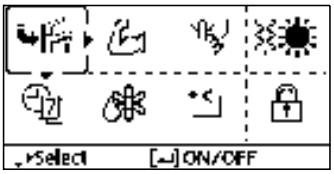
④ **Once the time is set, time and day will appear on the display even if the Remote Controller is turned OFF.**



5.1.3 Quick Menu

After the initial settings have been completed, you can select a quick menu from the following options and edit the setting.

① **Press ⊞ to display the quick menu.**



Force DHW | Powerful | Quiet | Force Heater

Weekly Timer | Force Defrost | Error Reset | R/C Lock

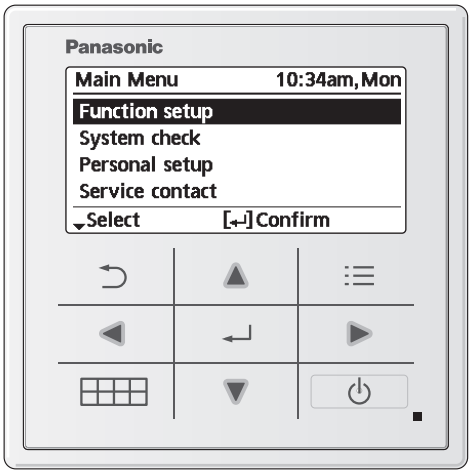
② **Use ▲ ▼ ◀ ▶ to select menu.**

③ **Press ↵ to turn on/off the select menu.**

5.1.4 Menus For user

Select menus and determine settings according to the system available in the household. All initial settings must be done by an authorised dealer or a specialist. It is recommended that all alterations of the initial settings are also done by an authorised dealer or a specialist.

- After initial installation, you may manually adjust the settings.
- The initial setting remains active until the user changes it.
- The Remote Controller can be used for multiple installations.
- Ensure the operation indicator is OFF before setting.
- The system may not work properly if set wrongly. Please consult an authorised dealer.



To display <Main Menu>:

To select menu:

To confirm the selected content:

Menu	Default Setting	Setting Options / Display
1 Function setup		
1.1 > Weekly timer		
Once the weekly timer is set up, User can edit from Quick Menu. To set up to 6 patterns of operation on a daily basis. • Disabled if Heat-Cool SW is select "Yes" or if Force Heater is on.	**Timer setup** Select day of the week and set the patterns needed (Time / Operation ON/OFF / Mode) **Timer copy** Select day of the week	Weekly timer 10:34am,Mon Sun Mon Tue Wed Thu Fri Sat 1. 8:00am ON 40℃ 2. 12:00pm ON 24/28℃ 40℃ 3. 1:00pm ON 12/10℃ ◆Day ▼Pattern [↵]Edit
1.2 > Holiday timer		
To save energy, a holiday period may be set to either turn OFF the system or lower the temperature during the period.	OFF	ON ▲ OFF
	> ON Holiday start and end. Date and time OFF or lowered temperature	Holiday: End 10:34am,Mon Year/Month/Day Hour : Min ▲ 2020 / 01 / 07 10 : 00 am ▼ ◆▶ Select [↵]Confirm
• Weekly timer setting may be temporarily disabled during Holiday timer setting but it will be restored once the Holiday timer is completed.		
1.3 > Quiet timer		
To operate quietly during the preset period. 6 patterns may be set. Level 0 means the mode is off.	Time to start Quiet : Date and time Level of quietness: 0 ~ 3	Quiet 10:34am,Mon Pattern Time Level 1 8:00am 0 2 5:00pm 1 3 11:00pm 3 ▼Select [↵]Edit
1.4 > Room heater		
To set the room heater ON or OFF.	OFF	ON ▲ OFF
1.5 > Tank heater		
To set the tank heater ON or OFF.	OFF	ON ▲ OFF
• Available only if connected to the tank.		
1.6 > Sterilization		
To set the auto sterilization ON or OFF.	OFF	ON ▲ OFF
• Available only if connected to the tank. • Do not use the system during sterilization in order to prevent scalding with hot water, or overheating of shower. • Ask an authorised dealer to determine the level of sterilization function field settings according to the local laws and regulations.		

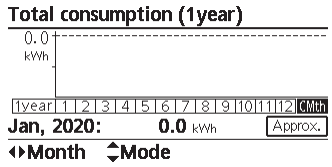
2 System check

2.1 > Energy monitor

Present or historical chart of energy consumption, generation or COP.

Present
Select and retrieve

Historical chart
Select and retrieve

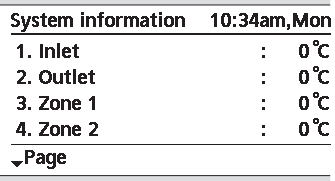


- COP= Coefficient of Performance.
- For historical chart, the period is selected from 1 day/1 week/1year.
- Energy consumption (kWh) of heating, *1, *2 cooling, tank and total may be retrieved.
- The total power consumption is an estimated value based on AC 230 V and may differ from value measured by precise equipment.

2.2 > System information

Shows all system information in each area.

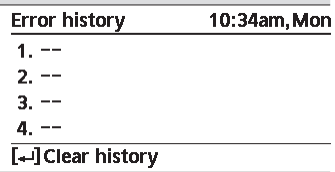
Actual system information of 10 items: Inlet / Outlet / Zone 1 / Zone 2 / Tank / Buffer tank / Solar / Pool / COMP frequency / Pump flowrate
Select and retrieve



2.3 > Error history

- Refer to Troubleshooting for error codes.
- The most recent error code is displayed at the top.

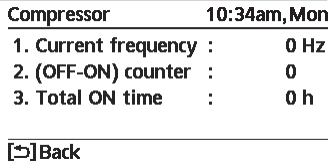
Select and retrieve



2.4 > Compressor

Shows the compressor performance.

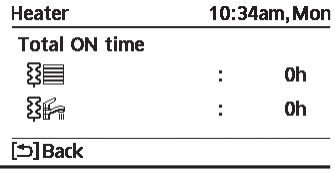
Select and retrieve



2.5 > Heater

Total hours of ON time for Room heater/Tank heater.

Select and retrieve



3 Personal setup

3.1 > Touch sound

Turns the operation sound ON/ OFF.

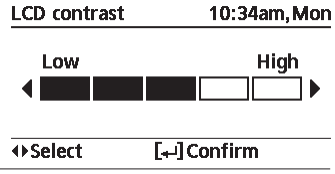
ON



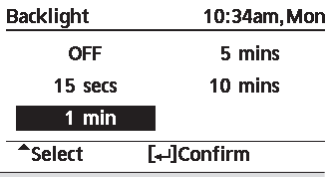
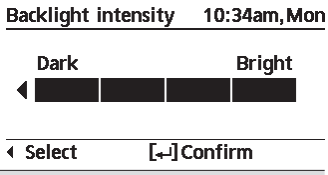
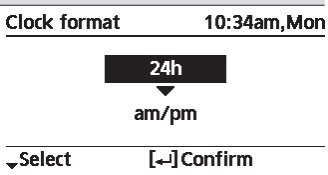
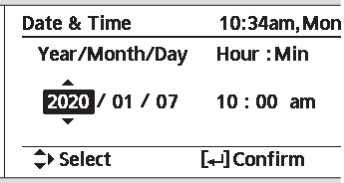
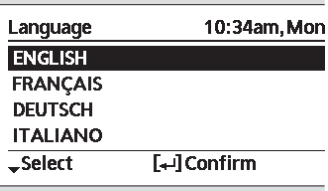
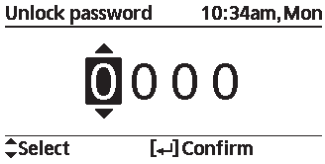
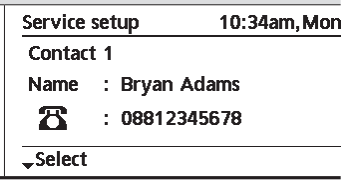
3.2 > LCD contrast

Sets the screen contrast.

3



*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

Menu	Default Setting	Setting Options / Display
3.3 > Backlight		
Sets the duration of screen backlight.	1 min	
3.4 > Backlight intensity		
Sets screen backlight brightness.	4	
3.5 > Clock format		
Sets the type of clock display.	24h	
3.6 > Date & Time		
Sets the present date and time.	Year / Month / Day / Hour / Min	
3.7 > Language		
Sets the display language for the top screen. • For Greek, please refer to the English version.	ENGLISH / FRANÇAIS / DEUTSCH / ITALIANO / ESPAÑOL / DANISH / SWEDISH / NORWEGIAN / POLISH / CZECH / NEDERLANDS / TÜRKÇE / SUOMI / MAGYAR / SLOVENŠČINA / HRVATSKI / LIETUVIŲ	
3.8 > Unlock password		
4 digit password for all the settings.	0000	
4 Service contact		
4.1 > Contact 1 / Contact 2		
Preset contact number for installer.	Select and retrieve	

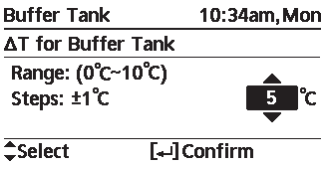
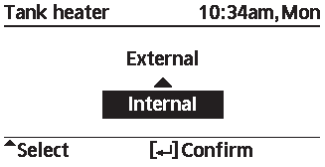
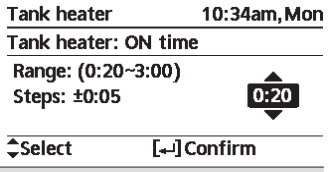
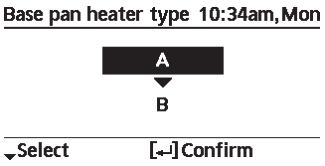
5.1.5 Menus For installer

Menu	Default Setting	Setting Options / Display
5 Installer setup > System setup		
5.1 > Optional PCB connectivity		
To connect to the external PCB required for servicing.	No	Yes ▲ No

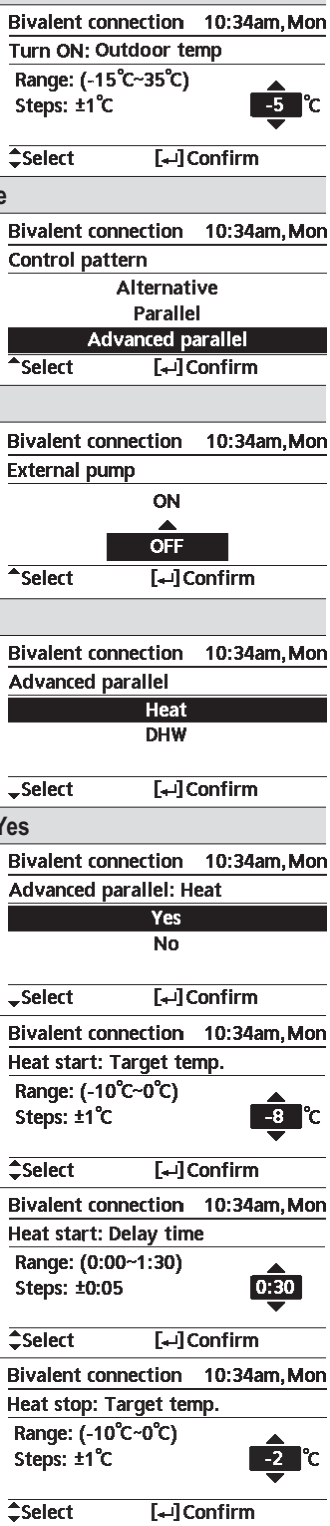
• If the external PCB is connected (optional), the system will have following additional functions:

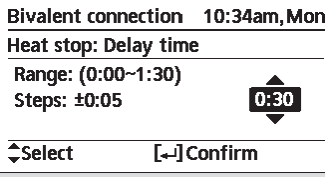
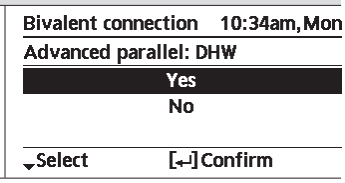
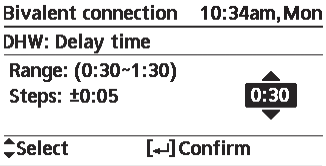
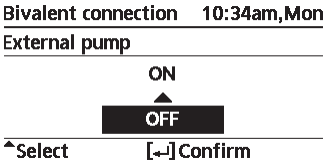
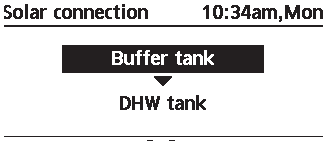
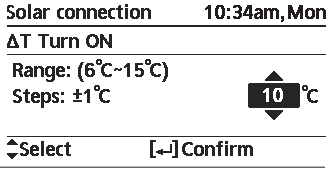
① Buffer tank connection and control over its function and temperature.
② Control over 2 zones (including the swimming pool and the function to heat water in it).
③ Solar function (the solar thermal panels connected to either the DHW (Domestic Hot Water) Tank or the Buffer Tank.
• DHW is not applicable for WH-ADC *models.
④ External compressor switch.
⑤ External error signal.
⑥ SG ready control.
⑦ Demand control.
⑧ Heat-Cool SW

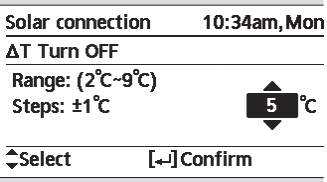
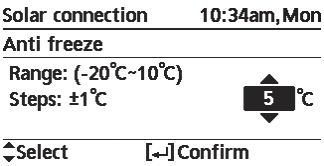
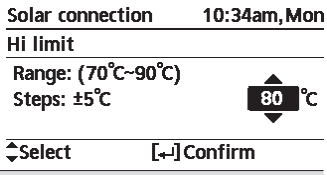
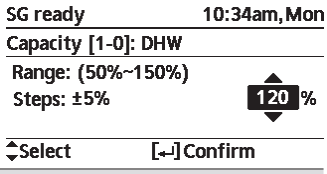
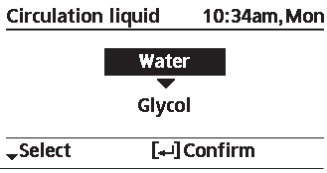
5.2 > Zone & Sensor		
To select the sensors and to select either 1 zone or 2 zone system.	**Zone** • After selecting 1 or 2 zone system, proceed to the selection of room or swimming pool. • If the swimming pool is selected, the temperature must be selected for △T temperature between 0°C ~ 10°C.	Zone & Sensor 10:34am,Mon Zone **1 Zone system** 2 Zones system ▼Select [↵]Confirm
	Sensor * For room thermostat, there is a further selection of external or internal.	Zone & Sensor 10:34am,Mon Sensor **Water temperature** Room thermostat Room thermistor ▼Select [↵]Confirm
5.3 > Heater capacity		
To reduce the heater power if unnecessary.* 3 kW / 6 kW / 9kW * Options of kW vary depending on the model.		Heater capacity 10:34am,Mon **3 kW** [↵]Confirm
5.4 > Anti freezing		
To activate or deactivate the water freeze prevention when the system is OFF	Yes	**Yes** ▼ No
5.5 > Tank connection		
To connect tank to the system.	No	Yes ▲ **No**
5.6 > DHW capacity		
To select tank heating capacity to variable or standard. Variable capacity heat up tank with fast mode and keep the tank temperature with efficient mode. While standard capacity heat up tank with rated heating capacity. * This option is available if Tank connection is selected (YES).	Variable	**Variable** ▼ **Standard**

Menu	Default Setting	Setting Options / Display
5.7 > Buffer tank connection		
To connect tank to the system and if selected YES, to set △T temperature. • The optional PCB connectivity must be selected YES to enable the function. • If the optional PCB connectivity is not selected, the function will not appear on the display.	No	
	> Yes	
	5°C	Set △T for Buffer Tank 
5.8 > Tank heater		
To select external or internal tank heater and if External is selected, set a timer for the heater to come on. * This option is available if Tank connection is selected (YES).	Internal	
	> External	
	0:20	Tank heater ON time set. 
5.9 > Base pan heater		
To select whether or not optional base pan heater is connected. * Type A - The base pan heater activates only during deice operation. * Type B - The base pan heater activates when outdoor ambient temperature is 5°C or lower.	No	
	> Yes	
	A	Set base pan heater type*. 
5.10 > Alternative outdoor sensor		
To select an alternative outdoor sensor.	No	
5.11 > Bivalent connection		
To select to enable or disable bivalent connection.	No	
> Yes		
To select either auto control pattern or SG ready input control pattern. * This selection only display to select when optional pcb connection set to Yes.	Auto	

Menu	Default Setting	Setting Options / Display
To select a bivalent connection to allow an additional heat source such as a boiler to heat-up the buffer tank and domestic hot water tank when heatpump capacity is insufficient at low outdoor temperature. The bivalent feature can be set-up either in alternative mode (heatpump and boiler operate alternately), or in parallel mode (both heatpump and boiler operate simultaneously), or in advance parallel mode (heatpump operates and boiler turns on for buffer-tank and/or domestic hot water depending on the control pattern setting options).	**> Yes > Auto**	
	-5°C	Set outdoor temperature for turn ON Bivalent connection.
	Yes > After selecting the outdoor temperature	
	Control pattern Alternative / Parallel / Advanced parallel • Select advanced parallel for bivalent use of the tanks.	
	Control pattern > Alternative	
	OFF	Option to set external pump either ON or OFF during bivalent operation. Set to ON if system is simple bivalent connection.
	Control pattern > Advanced parallel	
	Heat • "Heat" implies Buffer Tank and "DHW" implies Domestic Hot Water Tank.	Selection of the tank
	Control pattern > Advanced parallel > Heat > Yes	
	• Buffer Tank is activated only after selecting "Yes".	
	-8°C	Set the temperature threshold to start the bivalent heat source.
	0:30	Delay timer to start the bivalent heat source (in hour and minutes).
	-2°C	Set the temperature threshold to stop the bivalent heat source.

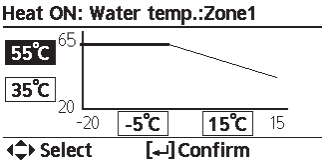
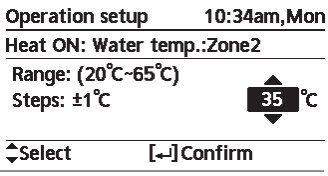
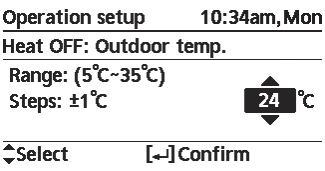
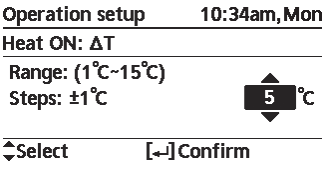
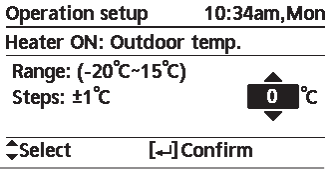


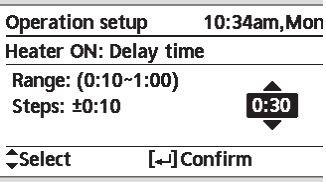
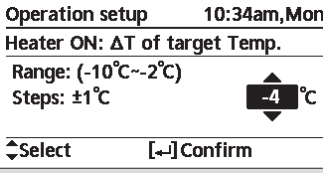
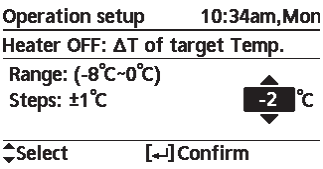
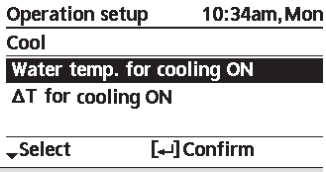
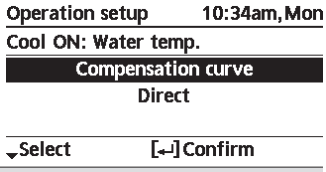
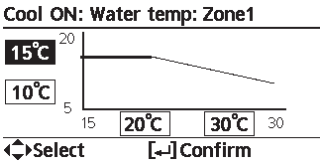
Menu	Default Setting	Setting Options / Display
	0:30	Delay timer to stop the bivalent heat source (in hour and minutes). 
	Control pattern > Advanced parallel > DHW > Yes	
	• DHW Tank is activated only after selecting "Yes".	
	0:30	Delay timer to start the bivalent heat source (in hour and minutes). 
SG ready input control for bivalent system follow below input condition. <table><tr><th colspan="2">SG signal</th><th>Operation pattern</th></tr><tr><th>Vcc-bit1</th><th>Vcc-bit2</th><th></th></tr><tr><td>Open</td><td>Open</td><td>Heat Pump OFF, Boiler OFF</td></tr><tr><td>Short</td><td>Open</td><td>Heat Pump ON, Boiler OFF</td></tr><tr><td>Open</td><td>Short</td><td>Heat Pump OFF, Boiler ON</td></tr><tr><td>Short</td><td>Short</td><td>Heat Pump ON, Boiler ON</td></tr></table>	**> Yes > SG ready**	
	OFF	Option to set external pump either ON or OFF during bivalent operation. Set to ON if system is simple bivalent connection. 
5.12 > External SW		
	No	
5.13 > Solar connection		
• The optional PCB connectivity must be selected YES to enable the function. • If the optional PCB connectivity is not selected, the function will not appear on the display. • DHW is not applicable for WH-ADC *models.	No	
	> Yes	
	Buffer tank	Selection of the tank 
	> Yes > After selecting the tank	
	10°C	Set △T ON temperature 

Menu	Default Setting	Setting Options / Display
	> Yes > After selecting the tank > △T ON temperature	
	5°C	Set △T OFF temperature 
	> Yes > After selecting the tank > △T ON temperature > △T OFF temperature	
	5°C	Set Antifreeze temperature 
	> Yes > After selecting the tank > △T ON temperature > △T OFF temperature > After setting the antifreeze temperature	
	80°C	Set Hi limit 
5.14 > External error signal		
	No	
5.15 > Demand control		
	No	
5.16 > SG ready		
	No	
	> Yes	
	120 %	Capacity (1) & (2) of DHW (in %), Heat (in %) and Cool (in°C) 
5.17 > External compressor SW		
	No	
5.18 > Circulation liquid		
To select whether to circulate water or glycol in the system.	Water	

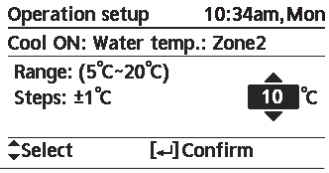
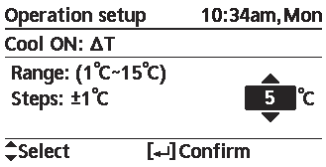
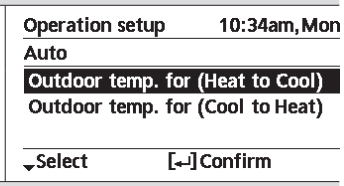
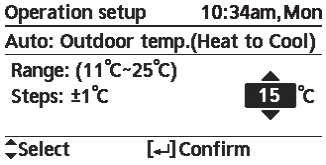
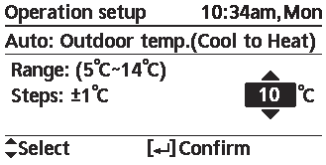
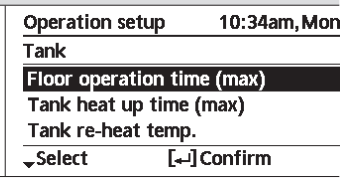
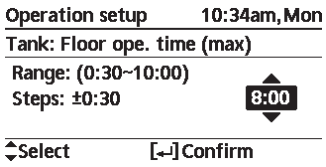
Menu	Default Setting	Setting Options / Display
5.19 > Heat-Cool SW		
	No	Yes ▲ No
5.20 > Force heater		
To turn on Force heater either manually (by default) or automatically.	Manual	Force heater 10:34am,Mon Auto ▲ Manual ▲Select [↵]Confirm
5.21 > Force defrost		
If auto selection is set, the unit will start defrost operation if long heating hour operate during low outdoor temperature.	Manual	Auto ▲ Manual
5.22 > Defrost signal		
To turn on defrost signal to stop fan coil during defrost operation. (If defrost signal set to yes, bivalent function will not available to use)	No	Yes ▲ No
5.23 > Pump flowrate		
To set variable flow pump control or fix pump duty control.	△T	ΔT ▼ Max. Duty
6 Installer setup > Operation setup		
To access to the four major functions or modes.	4 main modes Heat / *1, *2 Cool / *1, *2 Auto / Tank	Operation setup 10:34am,Mon Heat Cool Auto Tank ▼Select [↵]Confirm
6.1 > Heat		
To set various water & ambient temperatures for heating.	Water temp. for heating ON / Outdoor temp. for heating OFF / △T for heating ON / Heater ON/OFF	Operation setup 10:34am,Mon Heat Water temp. for heating ON Outdoor temp. for heating OFF ΔT for heating ON ▼Select [↵]Confirm
	> Water temp. for heating ON	
	Compensation curve	Heating ON temperatures in compensation curve or direct input.	Operation setup 10:34am,Mon Heat ON: Water temp. Compensation curve Direct ▼Select [↵]Confirm

*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

Menu	Default Setting	Setting Options / Display
	> Water temp. for heating ON > Compensation curve	
	X axis: -5°C, 15°C Y axis: 55°C, 35°C	Input the 4 temperature points (2 on horizontal X axis, 2 on vertical Y axis).	
	• Temperature range: X axis: -20°C ~ 15°C, Y axis: See below • Temperature range for the Y axis input: 1. WH-MXC model: 20°C ~ 65°C Regardless of the above setting, there is a limit to the water set temperature. Refer to the operating conditions on page 3. • If 2 zone system is selected, the 4 temperature points must also be input for Zone 2. • "Zone 1" and "Zone 2" will not appear on the display if only 1 zone system.	
	> Water temp. for heating ON > Direct	
	35°C	Temperature for heating ON	
	• Min. ~ Max. range is conditional as follows: 1. WH-MXC model: 20°C ~ 65°C Regardless of the above setting, there is a limit to the water set temperature. Refer to the operating conditions on page 3. • If 2 zone system is selected, temperature set point must input for Zone 2. • "Zone 1" and "Zone 2" will not appear on the display if only 1 zone system.	
	> Outdoor temp. for heating OFF	
	24°C	Temperature for heating OFF	
	> △T for heating ON	
	5°C	Set △T for heating ON. * This setting will not available to set when pump flowrate set to Max. duty.	
	> Heater ON/OFF	
	> Heater ON/OFF > Outdoor temp. for heater ON	
	0°C	Temperature for heater ON	

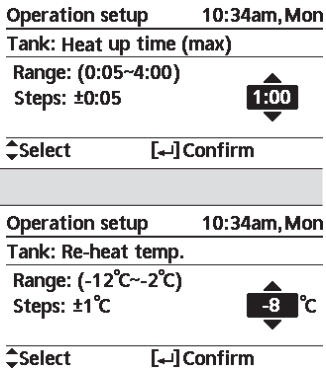
Menu	Default Setting	Setting Options / Display
	> Heater ON/OFF > Delay time for heater ON	
	0:30 min	Delay time for heater to turn on 
	> Heater ON/OFF > Water temperature for heater ON	
	-4°C	Setting of water temperature to turn on from water set temperature. 
	> Heater ON/OFF > Water temperature for heater OFF	
	-2°C	Setting of water temperature to turn off from water set temperature. 
6.2	**> *1, *2 Cool**	
To set various water & ambient temperatures for cooling.	Water temperatures for cooling ON and △T for cooling ON.	
	> Water temp. for cooling ON	
	Compensation curve	Cooling ON temperatures in compensation curve or direct input. 
	> Water temp. for cooling ON > Compensation curve	
	X axis: 20°C, 30°C Y axis: 15°C, 10°C	Input the 4 temperature points (2 on horizontal X axis, 2 on vertical Y axis) 
	• If 2 zone system is selected, the 4 temperature points must also be input for Zone 2. • "Zone 1" and "Zone 2" will not appear on the display if only 1 zone system.	

*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

Menu	Default Setting	Setting Options / Display
	> Water temp. for cooling ON > Direct	
	10°C	Set temperature for Cooling ON 
	• If 2 zone system is selected, temperature set point must input for Zone 2. • "Zone 1" and "Zone 2" will not appear on the display if only 1 zone system.	
	> △T for cooling ON	
	5°C	Set △T for cooling ON * This setting will not available to set when pump flowrate set to Max. duty. 
6.3	**> *1, *2 Auto**	
Automatic switch from Heat to Cool or Cool to Heat.	Outdoor temperatures for switching from Heat to Cool or Cool to Heat. Outdoor temp. for (Heat to Cool) / Outdoor temp. for (Cool to Heat)	
	> Outdoor temp. for (Heat to Cool)	
	15°C	Set outdoor temperature for switching from Heat to Cool. 
	> Outdoor temp. for (Cool to Heat)	
	10°C	Set outdoor temperature for switching from Cool to Heat. 
6.4	**> Tank**	
Setting functions for the tank. • Available only if connected to the tank.	Floor operation time (max) / Tank heat up time (max) / Tank re-heat temp. / Sterilization	
	• The display will show 3 functions at a time.	
	> Floor operation time (max)	
	8:00	Maximum time for floor operation (in hours and minutes) 

*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

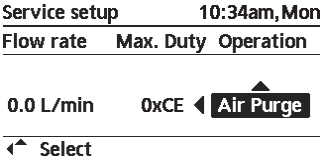
Menu	Default Setting	Setting Options / Display
	> Tank heat up time (max)	
	1:00	Maximum time for heating the tank (in hours and minutes)
	> Tank re-heat temp.	
	-8°C	Set temperature to perform reboil of tank water.
	> Sterilization	
	Monday	Sterilization may be set for 1 or more days of the week. Sun / Mon / Tue / Wed / Thu / Fri / Sat
	> Sterilization: Time	
	12:00	Time of the selected day(s) of the week to sterilize the tank 0:00 ~ 23:59
	> Sterilization: Boiling temp.	
	65°C	Set boiling temperatures for sterilize the tank.
	> Sterilization: Ope. time (max)	
	0:10	Set sterilizing time (in hours and minutes)

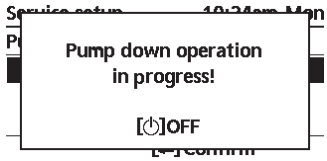
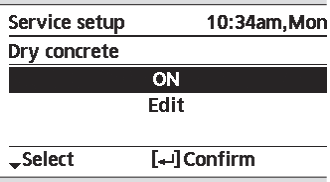
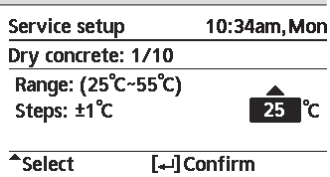
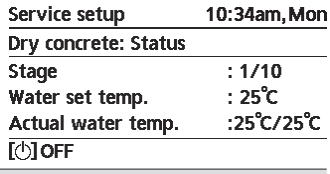
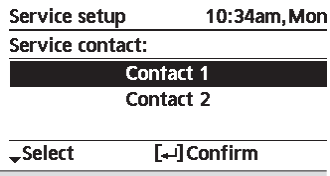
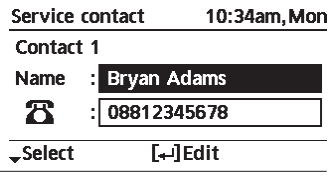
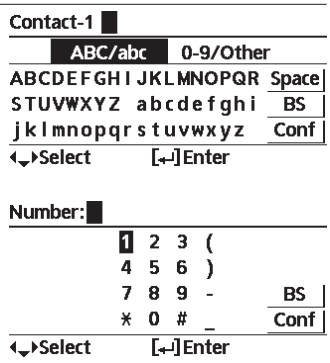


7 Installer setup > Service setup

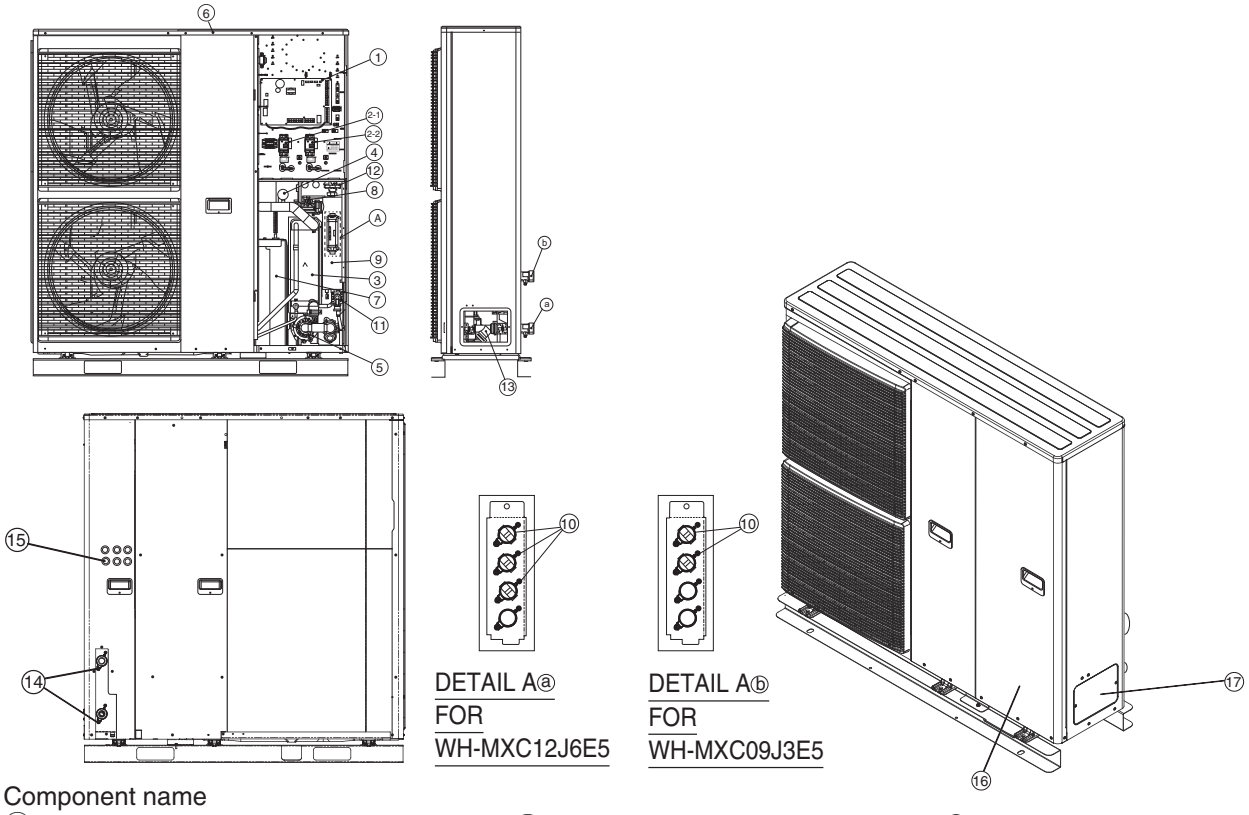
7.1 > Pump maximum speed

To set the maximum speed of the pump.	Setting the flow rate, max. duty and operation ON/OFF of the pump. Flow rate: XX:X L/min Max. Duty: 0x40 ~ 0xFE, Pump: ON/OFF/Air Purge	



Menu	Default Setting	Setting Options / Display
7.2 > Pump down		
To set the pump down operation.	**Pump down operation** ON	
7.3 > Dry concrete		
To dry the concrete (floor, walls, etc.) during construction. Do not use this menu for any other purposes and in period other than during construction	Edit to set the temperature of dry concrete. ON / Edit	
	> Edit	
	Stages: 1 Temperature: 25°C Heating temperature for drying the concrete. Select the desired stages: 1 ~ 10, range: 1 ~ 99	
	> ON	
	Confirm the setting temperatures of dry concrete for each stage.	
7.4 > Service contact		
To set up to 2 contact names and numbers for the User.	Service engineer's name and contact number. Contact 1 / Contact 2	
	> Contact 1 / Contact 2	
	Contact name or number. Name / phone icon	
	Input name and number Contact name: alphabet a ~ z. Contact number: 1 ~ 9	

5.1.6 Main Components



Component name

① PCB
②-1 Single phase RCCB (Main Power)
②-2 Single phase RCCB (Backup Heater)
③ Heat Exchanger
④ Water Pressure gauge
⑤ Water pump
⑥ Cabinet top plate
⑦ Expansion vessel (Not Visible)
⑧ Flow sensor
⑨ Heater assembly
⑩ Overload protector
⑪ Pressure relief valve
⑫ Air purge valve
⑬ Water Filter Set
⑭ Plug (2 pieces)
⑮ Bushing (6 pieces)
⑯ Cabinet front plate
⑰ Cover

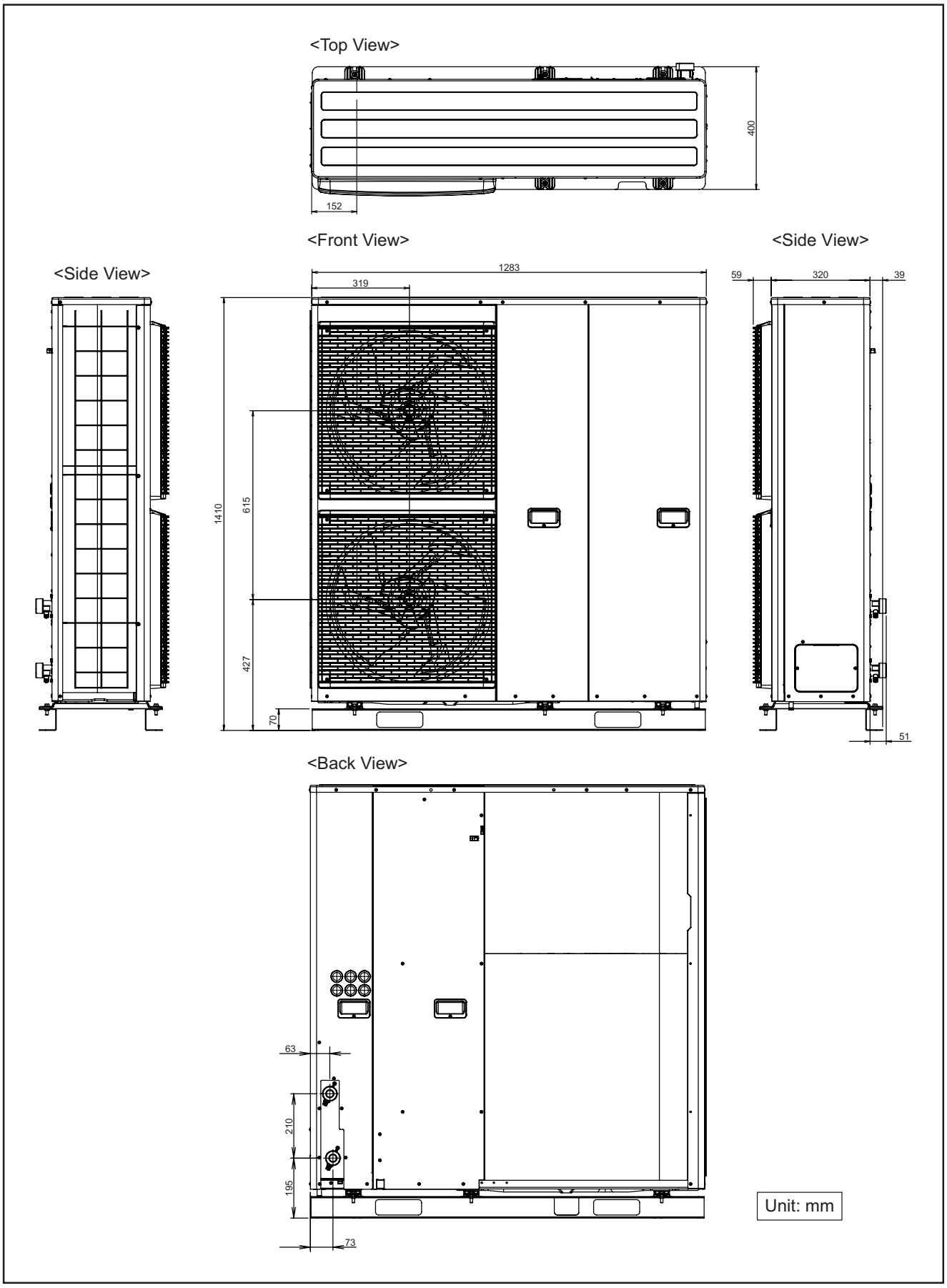
Connector name

ⓐ Water inlet
ⓑ Water outlet

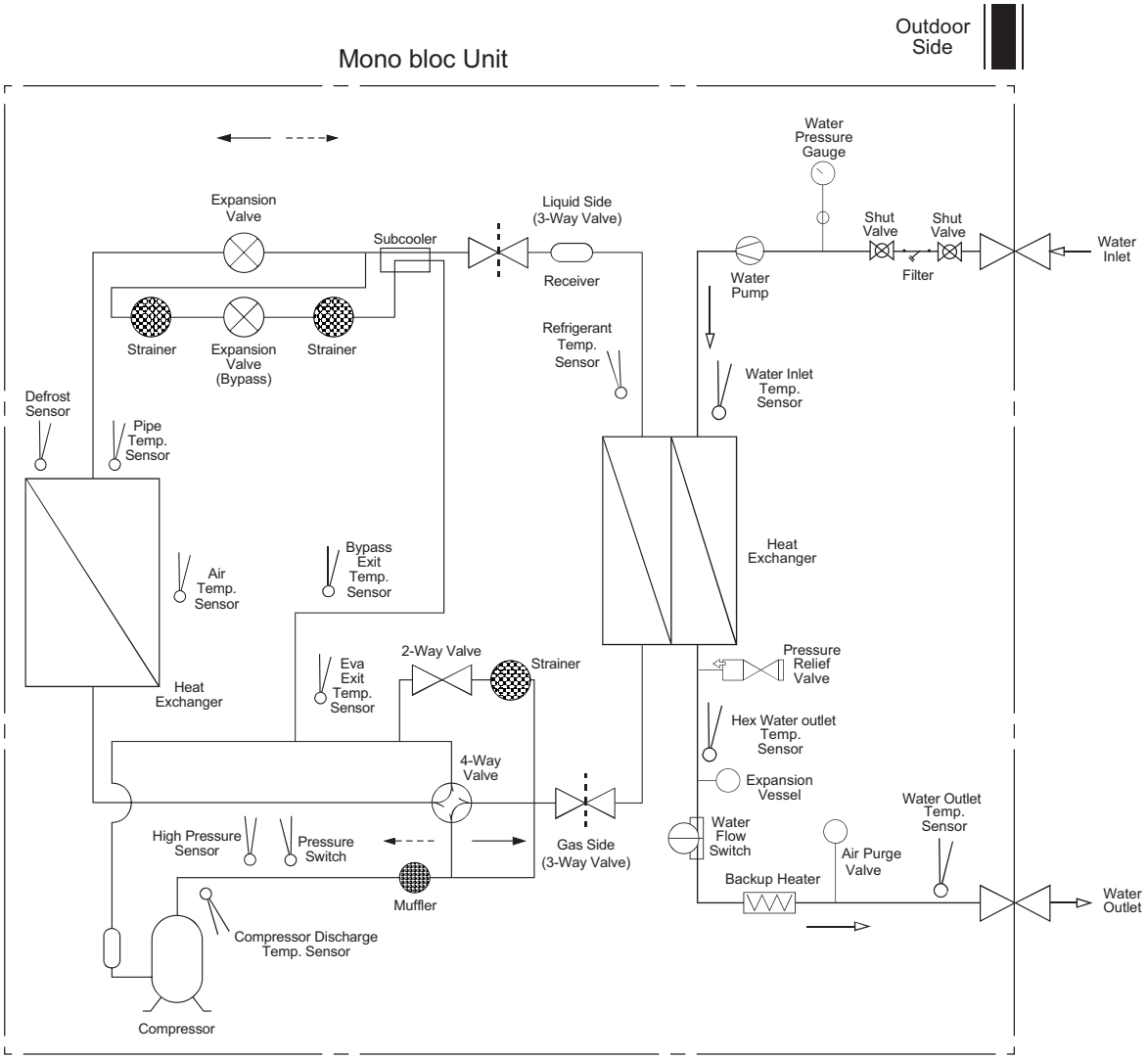
It is advisable to avoid more than 2 blockage directions. For better ventilation & multiple-outdoor installation, please consult authorized dealer/specialist.

6. Dimensions

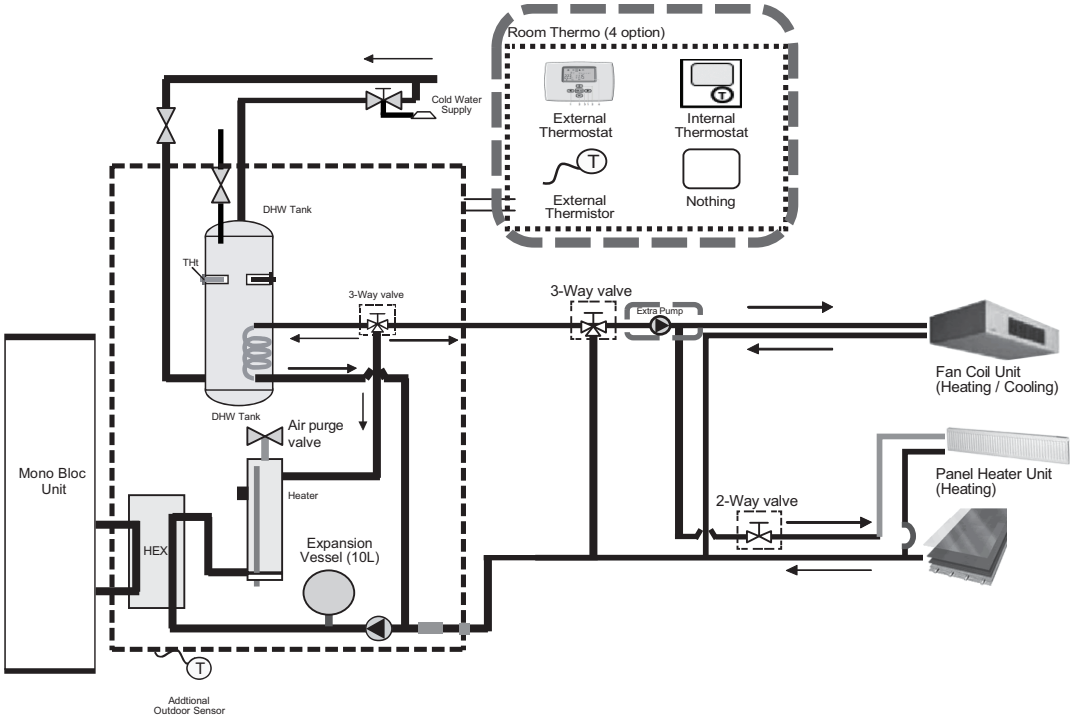
6.1 Mono Bloc Unit



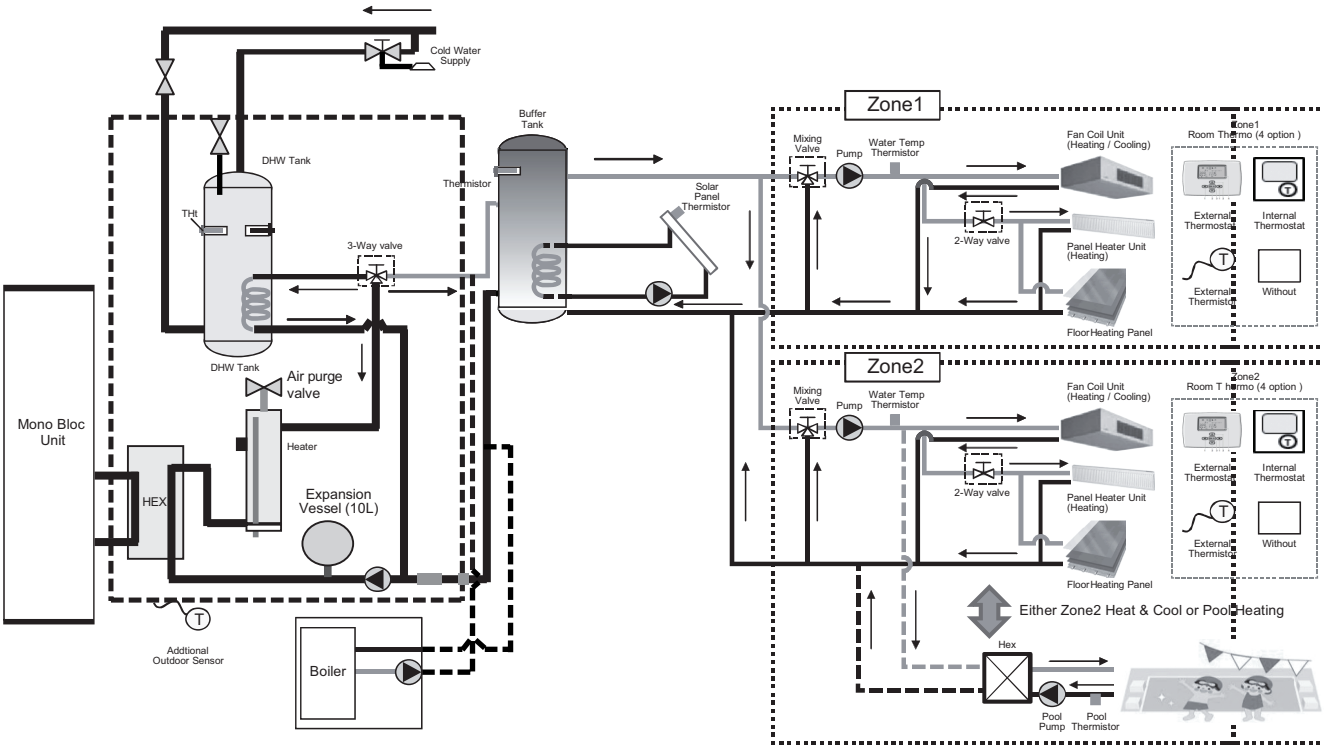
7. Refrigeration and Water Cycle Diagram



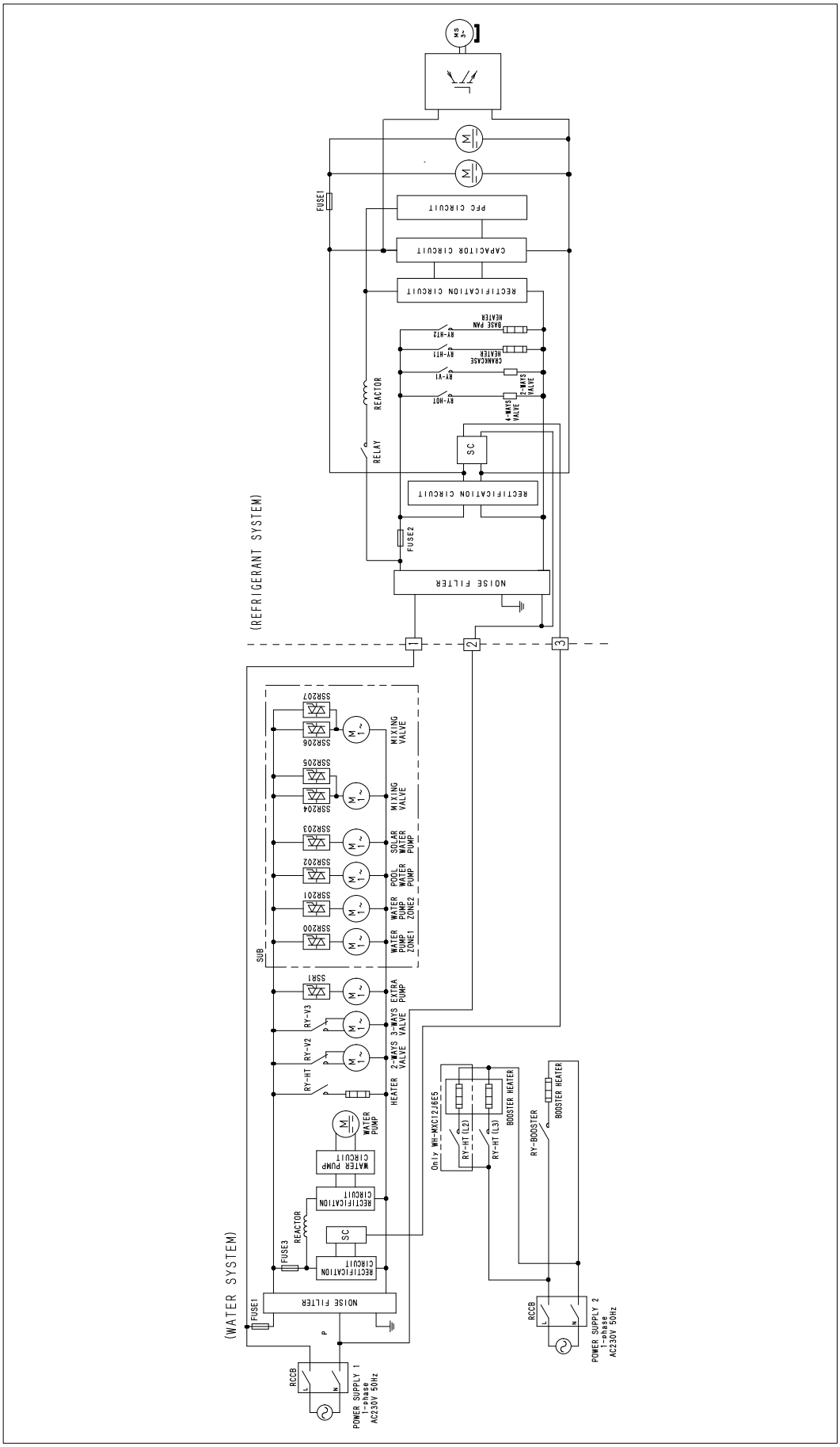
7.1 Standard System (Optional PCB not connected)



7.2 Advance System (Optional PCB connected)



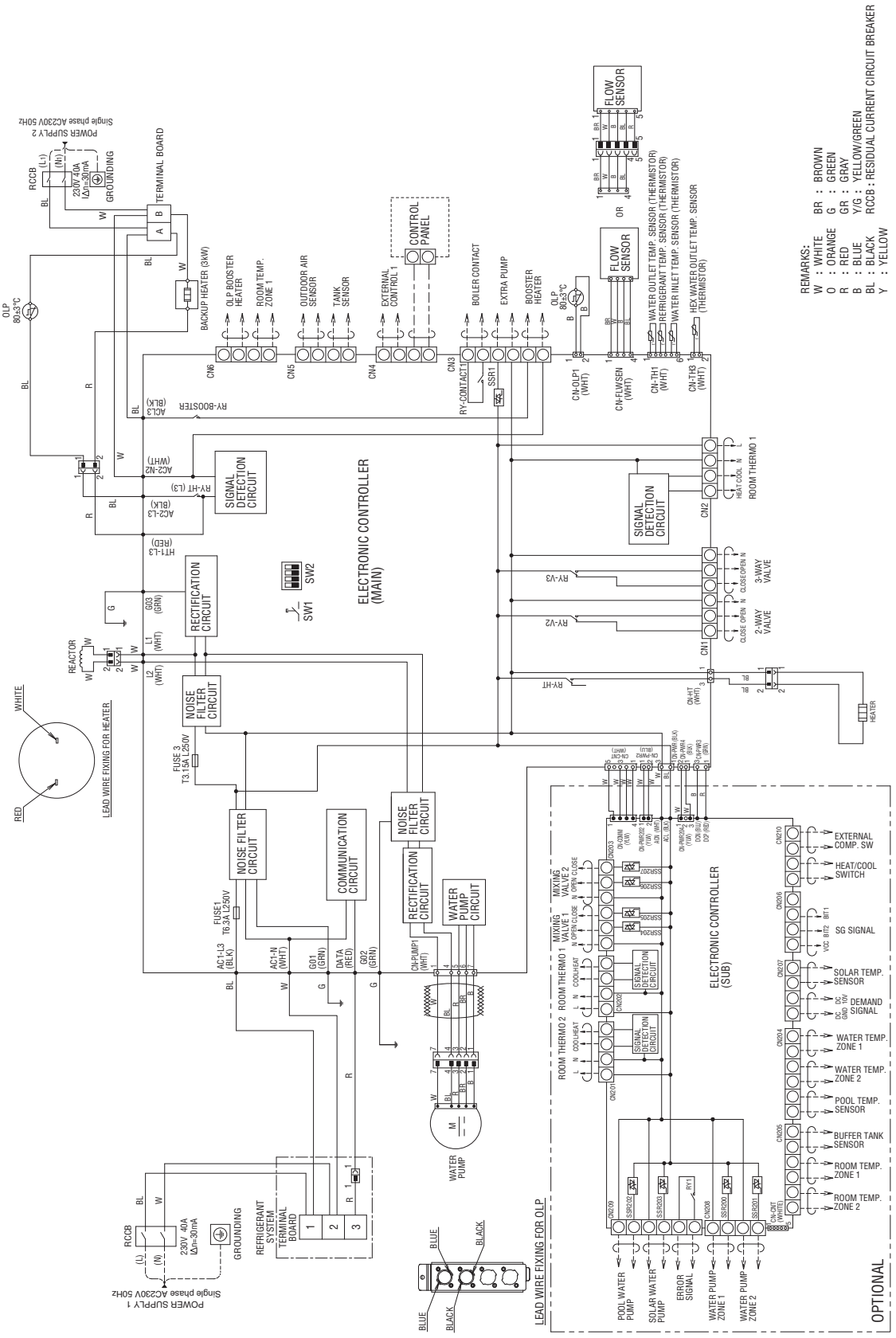
8. Block Diagram



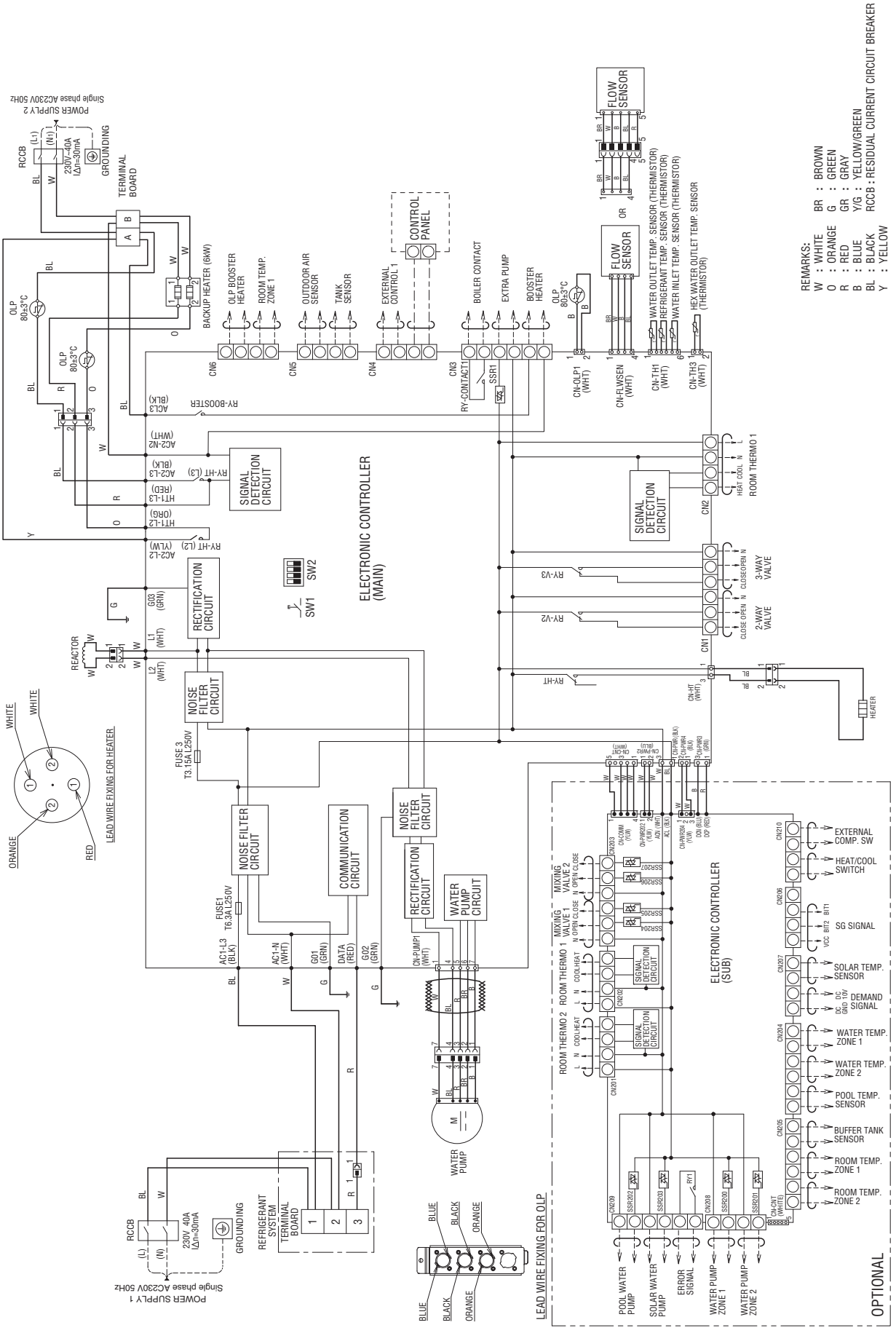
9. Wiring Connection Diagram

9.1 Water System

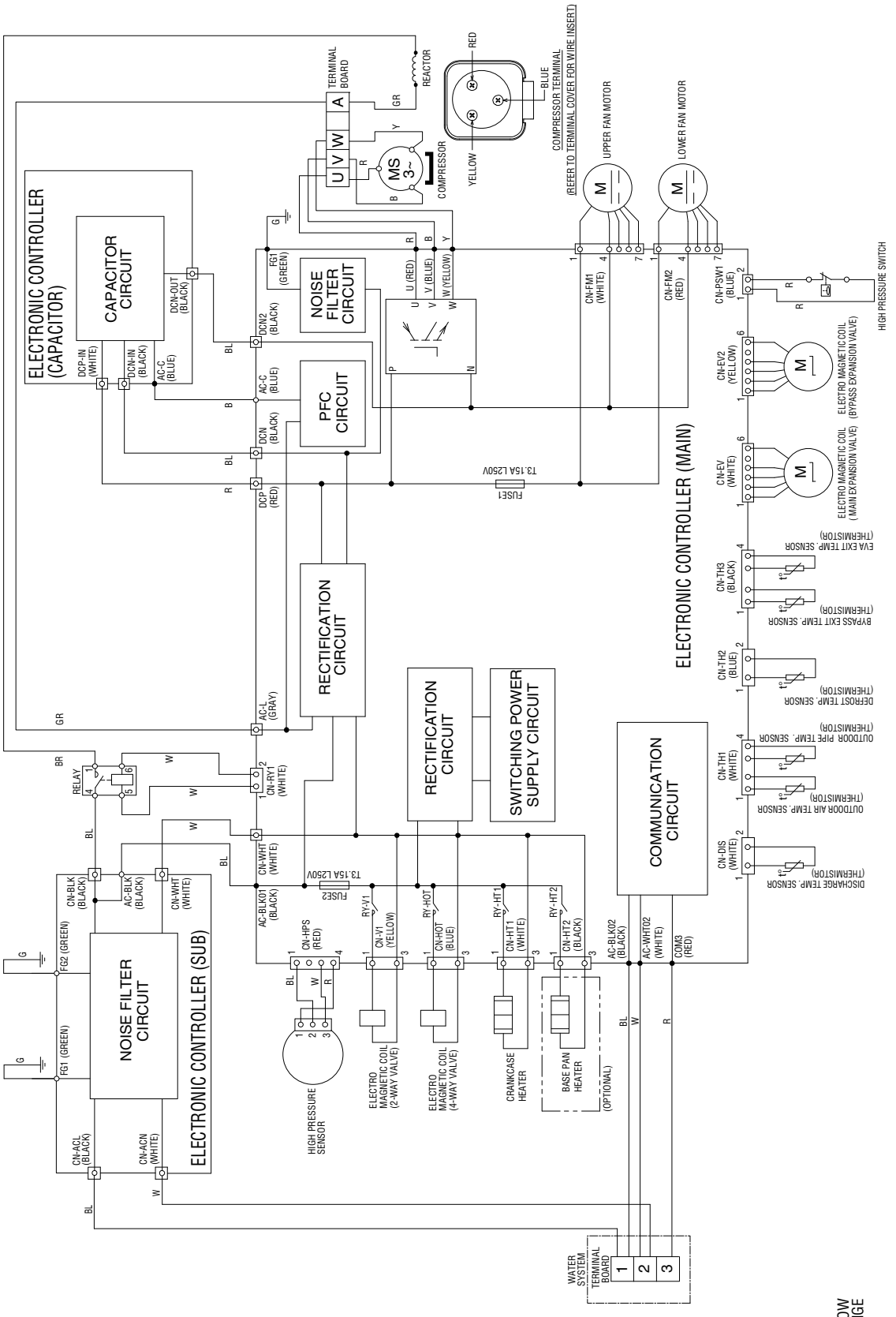
9.1.1 WH-MXC09J3E5



9.1.2 WH-MXC12J6E5



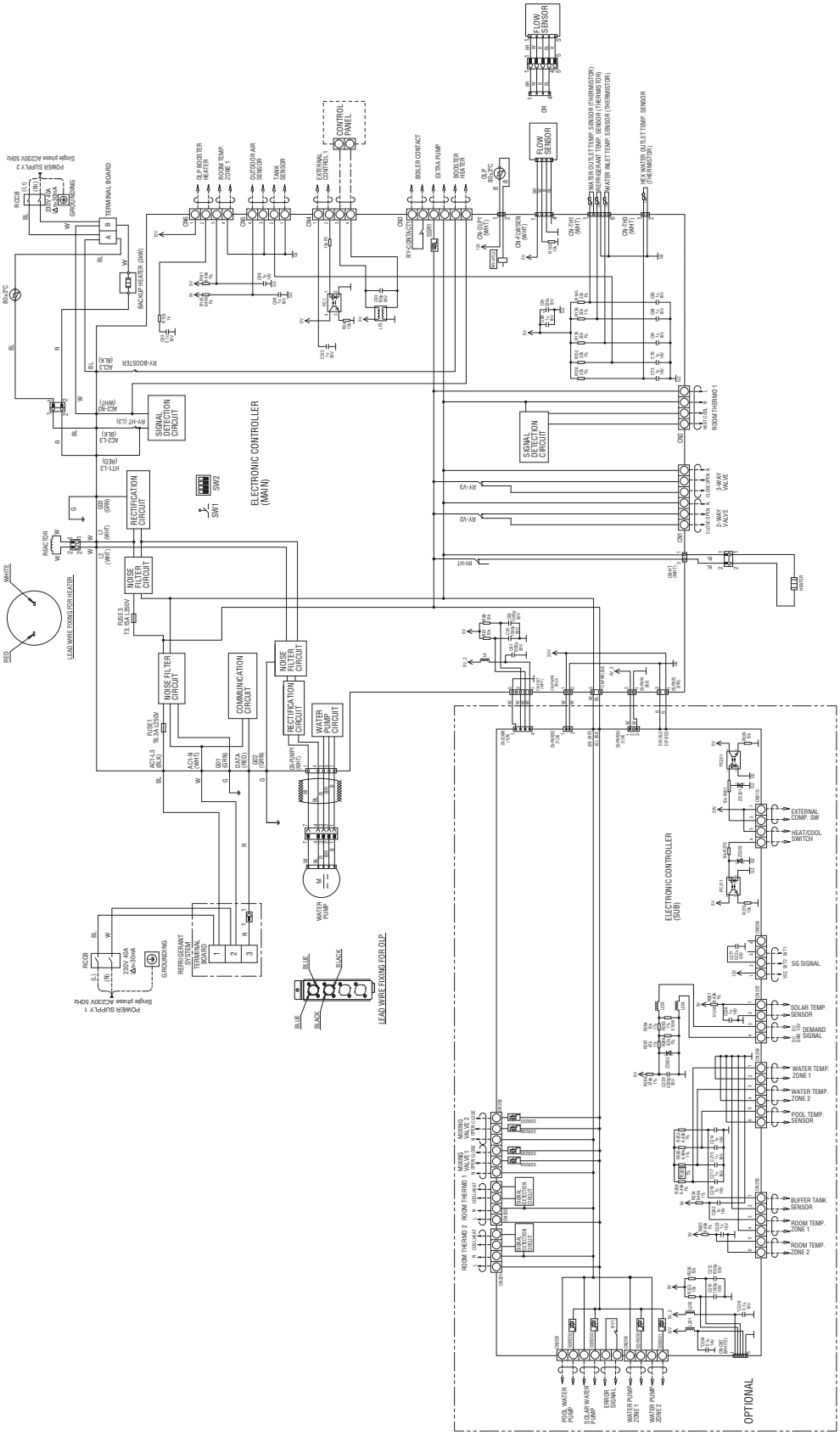
9.2 Refrigerant System



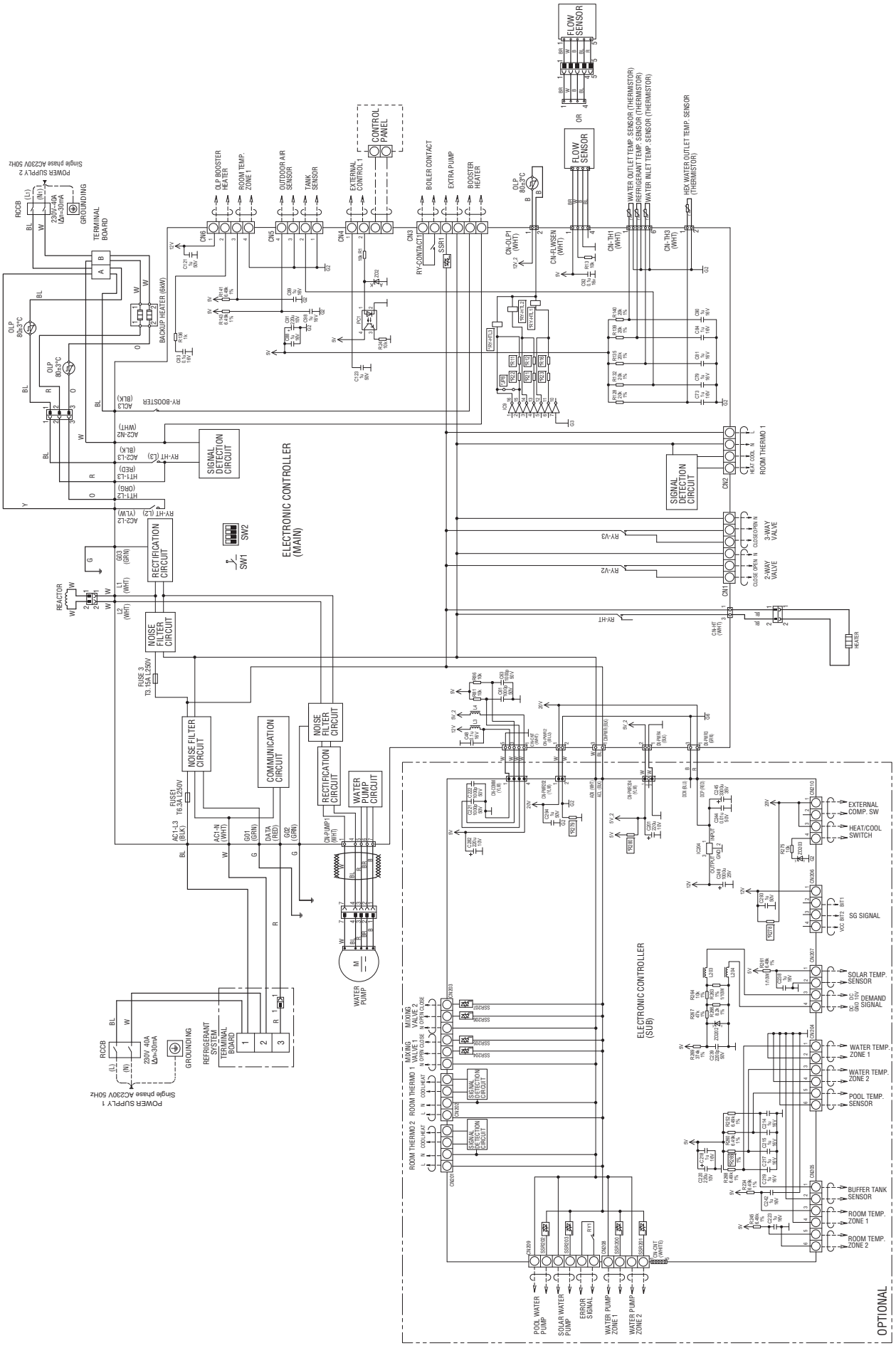
10. Electronic Circuit Diagram

10.1 Water System

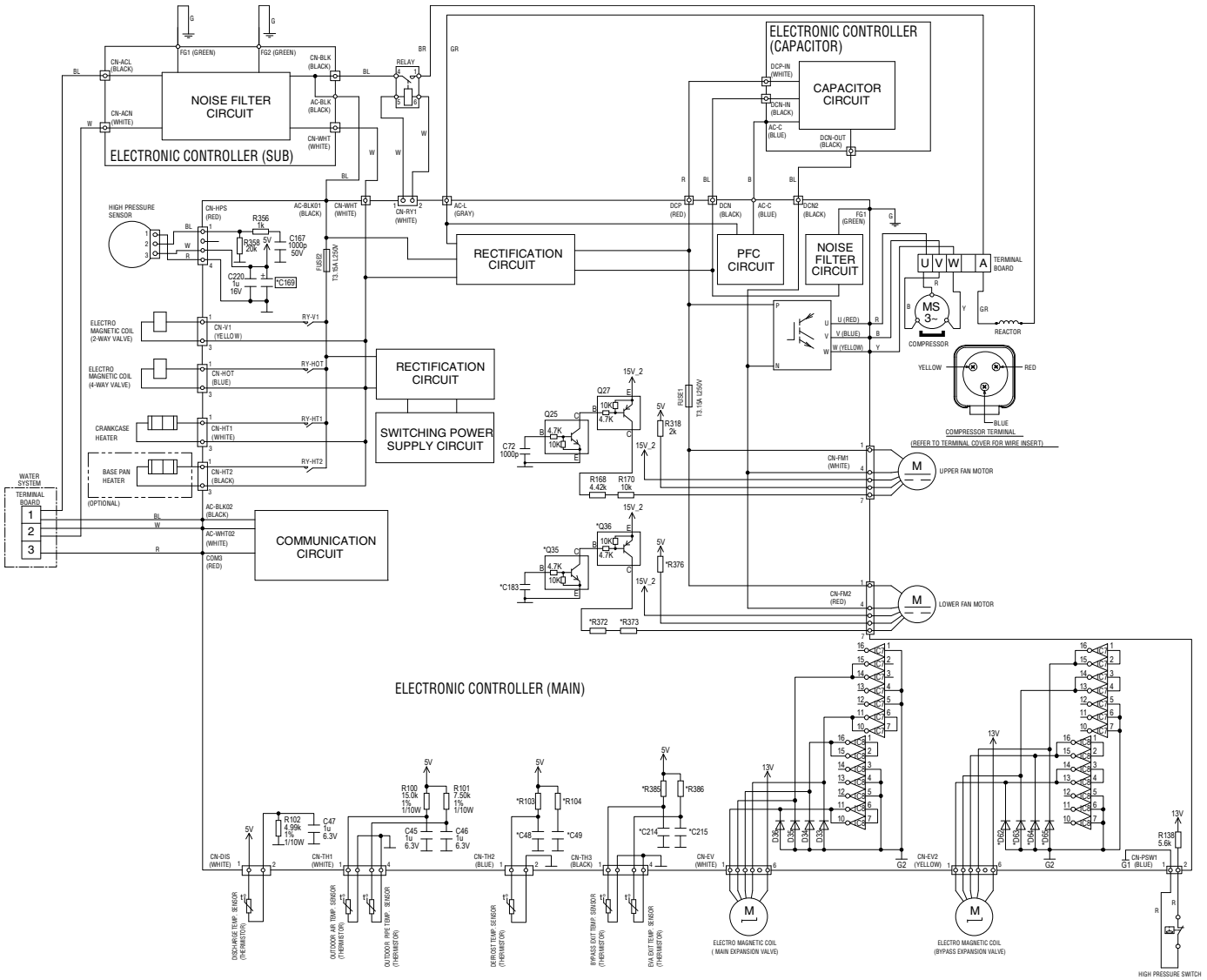
10.1.1 WH-MXC09J3E5

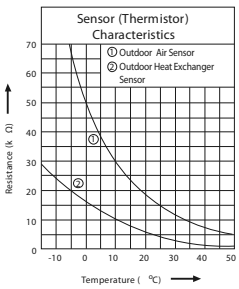


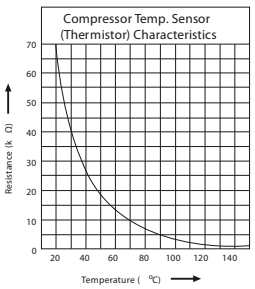
10.1.2 WH-MXC12J6E5



10.2 Refrigerant System



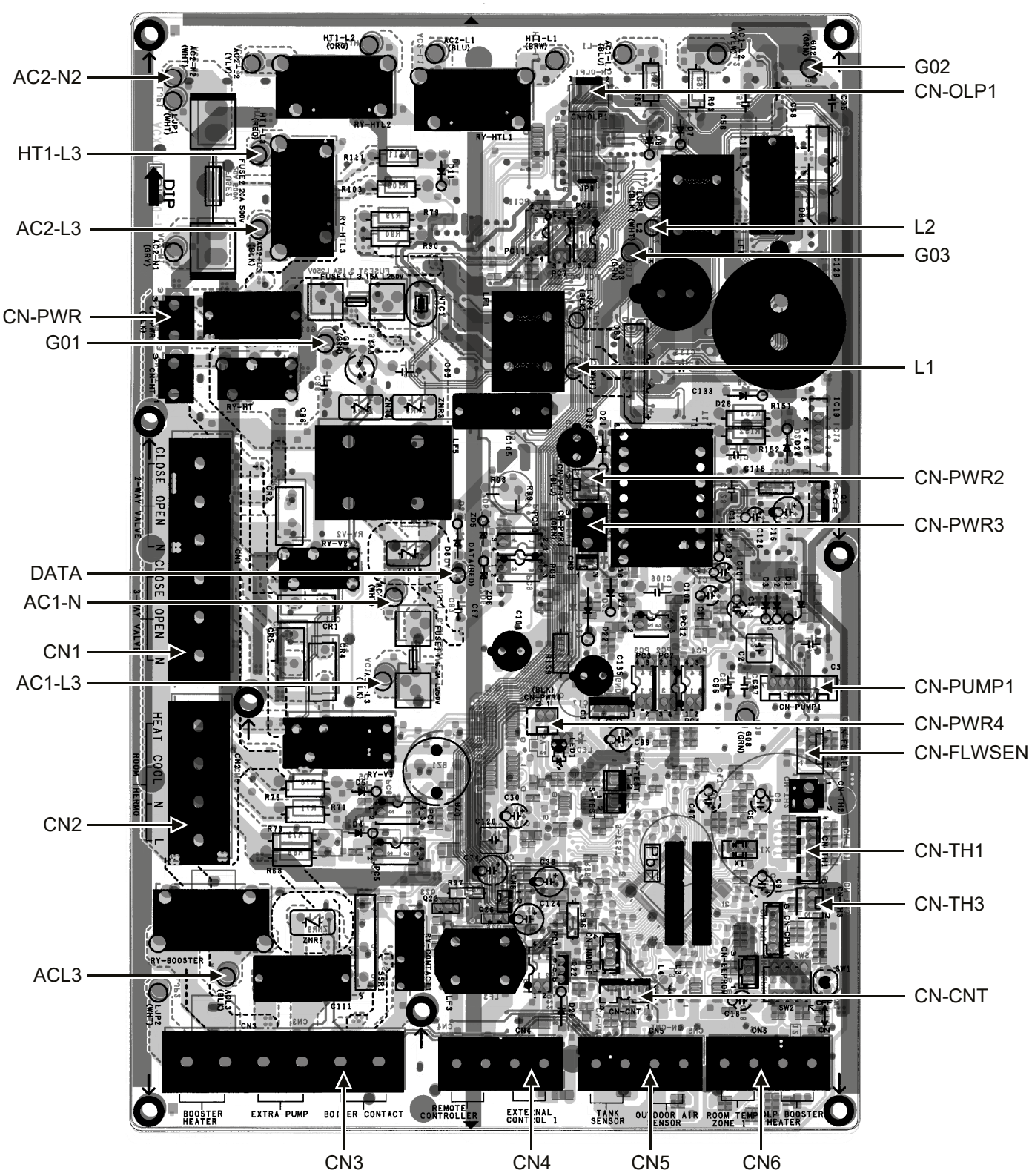




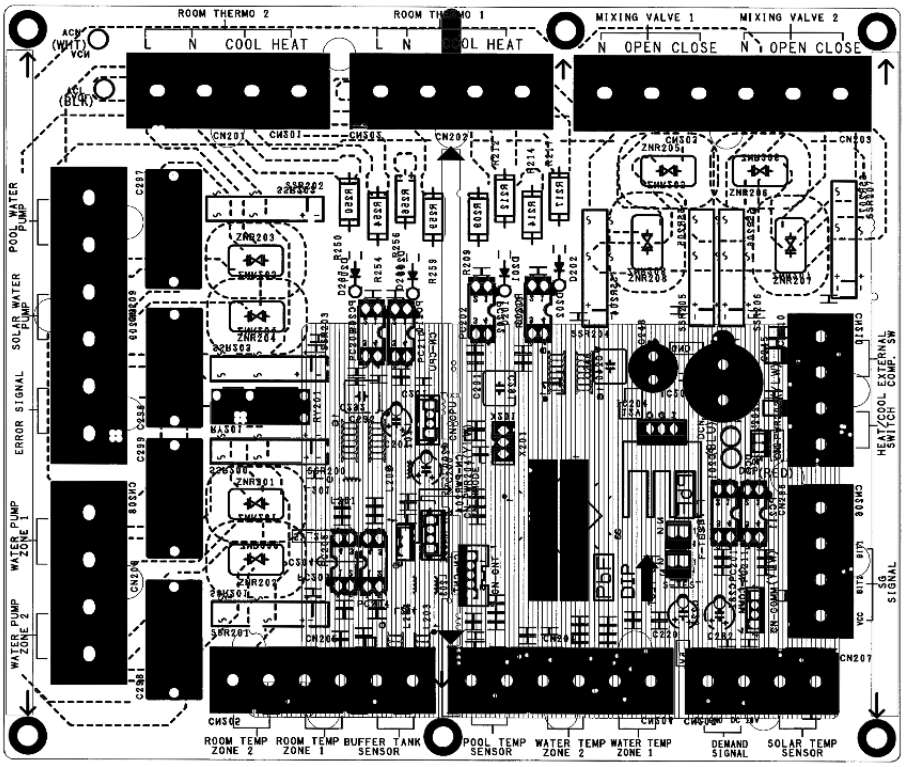
11. Printed Circuit Board

11.1 Water System

11.1.1 Main Printed Circuit Board

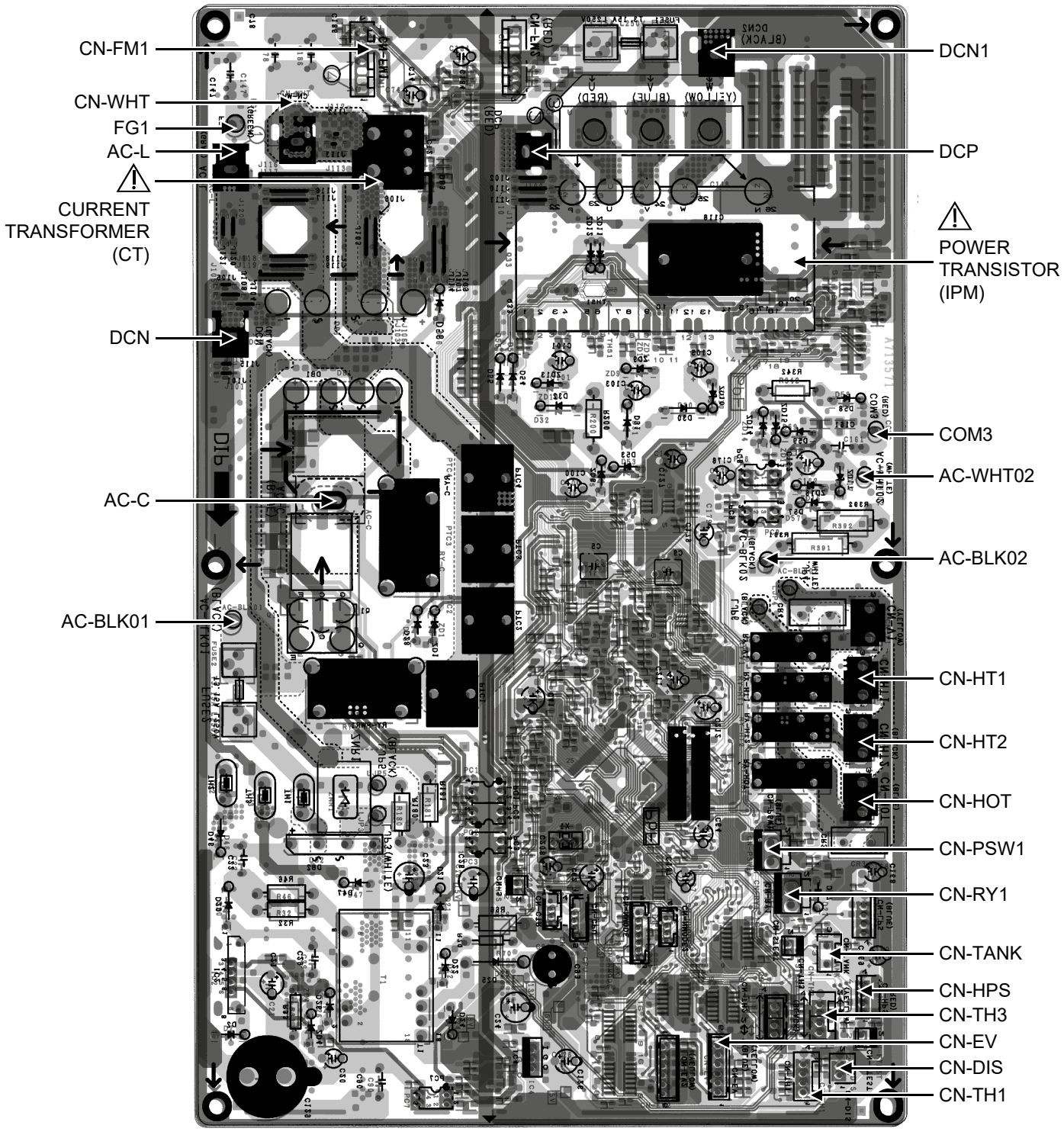


11.1.2 Optional Printed Circuit Board

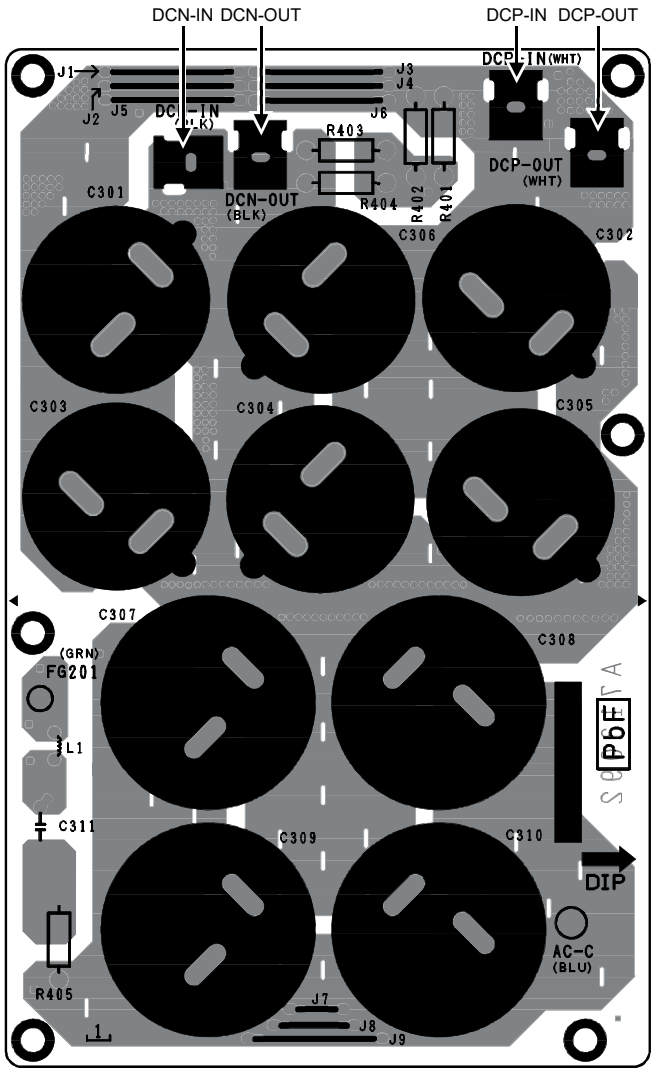


11.2 Refrigerant System

11.2.1 Main Printed Circuit Board



11.2.2 Capacitor Printed Circuit Board



11.2.3 Noise Filter Printed Circuit Board

